AutoAgentAI — Full Documentation

Combined export of the documentation site

2026-10-31

AutoAgentAI
Executive Summary
AutoAgentAI — Investor Pitch
AutoAgentAI — Technical Overview
AutoAgentAI — For Sales Managers
Vision
Problem & Motivation
Concepts & Glossary
Architecture
Customer Integration Layer
Insights Engine
Behaviour Models
Decision Intelligence
The AutoAgent Loop
Use Cases
On-Prem, Local AI & Hardware
Privacy, PII & Future-Proofing
Cloud vs On-Prem: The Risks
Local Models Are Enough
From Raw Data to a Decision: A Dealership Scenario
Reinventing the Salesperson's Role
Why Salespeople Can't Capture This Data
Capturing the Car Industry's Most Valuable Data
How Valuable This Data Is
Valuable Auto Data & Closing the Loop to the OEM
The Closed Loop: Impossible Before, Ubiquitous Elsewhere
Roadmap
Specifications
Figure Specs

About this document

Combined export of the AutoAgentAI documentation site, assembled into one file for review and editing. The source of truth remains the GitHub repo; edits here can be merged back. Regenerate with `make exports`.

AutoAgentAI

AutoAgent and Customer-Integrated Custom Insights into Purchase and Overall Behaviour and Decision Making.

One-line definition

AutoAgentAI is an **autonomous AI agent** that unifies a business's customer data and continuously produces **custom insights** into how customers **buy**, **behave**, and **decide** — then surfaces or acts on those insights to improve decisions.

Focus: automotive, on-prem, private

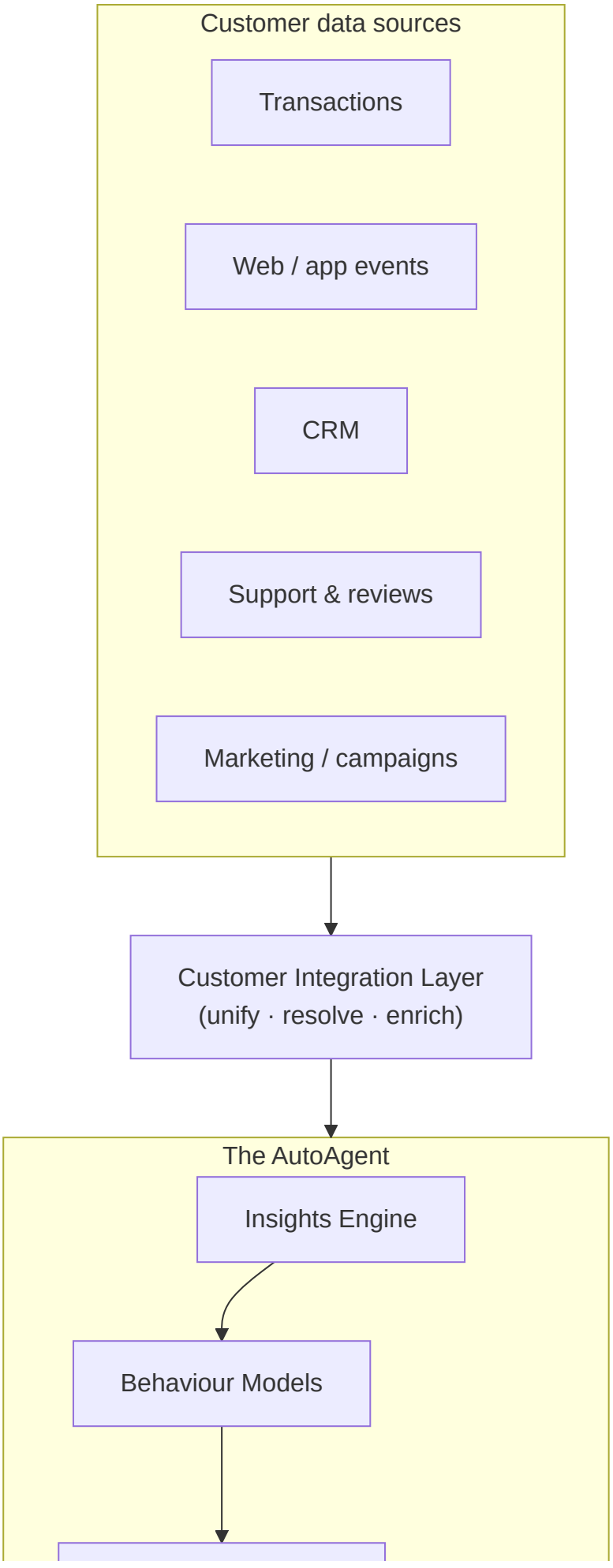
The beachhead is the **car industry** — dealerships and, through them, OEMs — where the buyer's pre-purchase decision behaviour is the most valuable, least-captured data there is. AutoAgentAI is built to run **on-prem on local AI models**, so sensitive customer data **never leaves the building**. See Deployment & Privacy and the Perspectives articles.

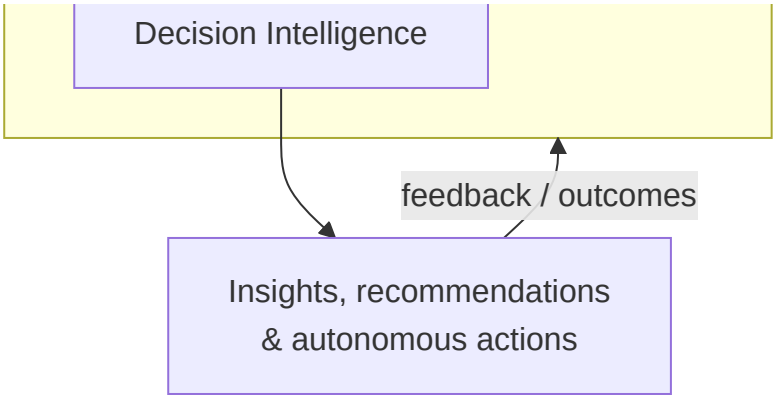
Why this exists

Businesses sit on fragmented customer data — transactions, web and app events, CRM records, support tickets, reviews, campaign responses. The signal about *why* customers buy, how they behave, and how they decide is spread across those silos and mostly goes unread. Static dashboards answer yesterday's questions and wait for a human to ask the next one.

AutoAgentAI replaces "dashboard you have to drive" with "**agent that drives itself**": it integrates the data, keeps looking, generates insights tailored to each business and customer, and turns them into recommended or automated actions.

The product in one picture


Customer data sources
Transactions
Web / app events
CRM
Support & reviews
Marketing / campaigns
Customer Integration Layer
(unify · resolve · enrich)
The AutoAgent
Insights Engine
Behaviour Models



What makes it distinct

Term in the name	What it means for the product
AutoAgent	Autonomous, goal-driven, continuous — not a static report. It decides what to look at next.
Customer-Integrated	Unifies every customer touchpoint into one resolved view per customer.
Custom Insights	Tailored to the specific business and customer — not generic benchmark metrics.
Purchase behaviour	What, when, how much, how often, and why customers buy (or don't).
Overall behaviour	Engagement, usage, support, sentiment — the whole relationship, not just the till.
Decision Making	Insights are framed as decisions: next-best-action, who to target, what to change.

Pick your lens

The same story, told for whoever's reading — start with whichever fits you:

- 📄 **Executive Summary** — the whole thing on one page (PDF).
- 💼 **Investor Pitch** — the opportunity, moat, model, and the ask.
- 🛠️ **Technical Overview** — architecture, on-prem stack, local models, security.
- 🚗 **For Sales Managers** — plain talk: what it does for your store.

How to read these docs

1. **Vision** and **Problem & Motivation** — the *why*.
2. **Concepts & Glossary** — the shared vocabulary.
3. **Architecture** — the *how*, pillar by pillar: Customer Integration · Insights Engine · Behaviour Models · Decision Making · The AutoAgent Loop.
4. **Use Cases** — made concrete.
5. **Deployment & Privacy** — on-prem/local AI, the four tiers, privacy & PII, and cloud vs on-prem risks.
6. **Perspectives** — the automotive thesis: the industry's most valuable data, the salesperson's changing role, and closing the loop to the OEM.
7. **Roadmap** and **Specifications** — the path to building.

> **Status**
>
> This is a **concept-and-documentation** repository. It organizes the thinking and hosts it, so that formal specs and implementation can follow. Everything here is a working draft meant to be challenged and refined.

Executive Summary

> **AutoAgentAI — one page**
>
> An autonomous AI agent that captures and uses the **most valuable, least-captured data in the car industry** — the buyer's real pre-purchase intent and decision behaviour — running **privately on-prem on local AI models**, and **closing the loop from the showroom floor to the OEM.**

The problem. Buying a car is a huge, considered, emotional decision, yet the data about *why* buyers choose, hesitate, or walk is almost entirely uncaptured — because it's structurally impossible for salespeople to capture. The industry runs on memory, sticky notes, lagging sales numbers, and survey-with-a-prize guesswork.

Why now. Three curves crossed at once: **agentic AI** can finally capture and reason over this signal; **open local models** make it private and cheap enough to run on-prem; and **tightening privacy law** makes sovereign, on-prem data the only durable way to hold it.

What it is. One agent across four jobs — unify customer data, model behaviour, generate custom insights, act on them — delivered through McDonald's-style kiosks and iPad nodes, running on the dealership's own hardware so **customer data never leaves the building.**

The wedge. This data scores at the top on every value driver: huge stakes per decision, a lifelong repeating relationship, genuine decision signal (not vanity metrics), near-total scarcity (nobody captures it), and value at **every** level — salesperson, dealer, group, and manufacturer.

The moat. The AI is commodity; the defensibility is **capture + privacy + the loop** — the on-floor footprint that captures the signal, the on-prem posture that can legally hold it, and the aggregated feedback loop to the OEM that compounds as it scales.

Business model. Four tiers as a pricing ladder: land with low-cost **Tier 0** cloud pilots, expand to on-prem **Tiers 1–3** (hardware + recurring license), multiply across dealer groups, and — longer term — monetize the **OEM-facing** buyer-intent insight product.

The impact on the floor. Fewer leads lost, reps who know the customer before they walk up, more trade-ins and service retention, and a salesperson's role that shifts from pressure-and-admin to **assurance and assistance** — a leaner, sharper team, each person amplified by the agent.

Status & ask. Concept and documentation complete; next is specifications plus a measured design-partner pilot. **We're seeking** a funding round *(amount to confirm)*, a **design-partner dealership**, and an **OEM introduction** to scope the feedback loop.

📄 **Download the one-page PDF** — print-ready, ready to send.

Full detail: Investor Pitch · Technical Overview · For Sales Managers · How Valuable This Data Is · Closing the Loop to the OEM.

AutoAgentAI — Investor Pitch

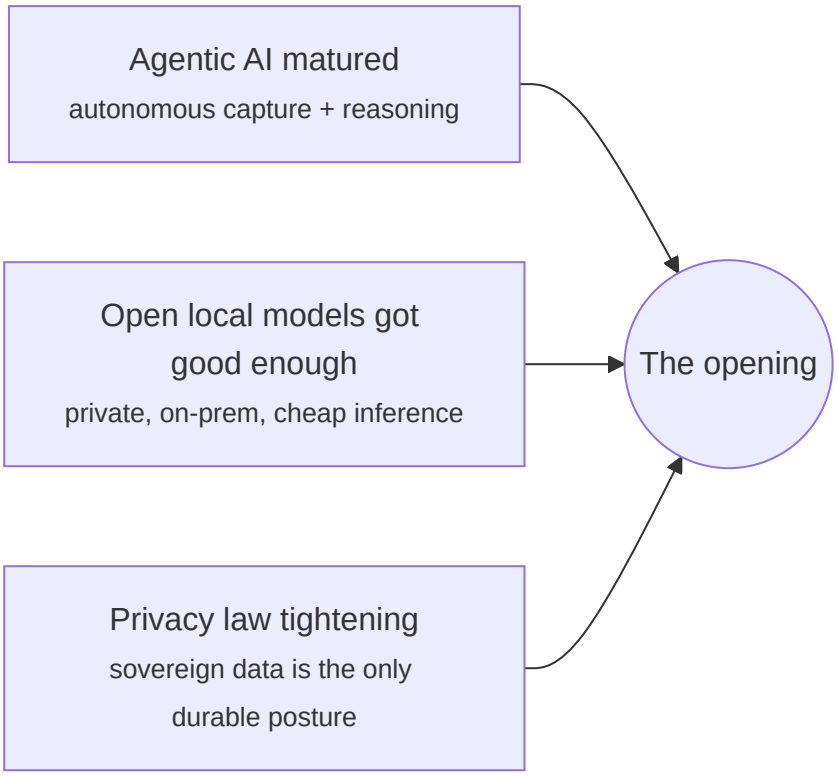
> **The one-liner**
>
> AutoAgentAI is an autonomous AI agent that captures and uses the **most valuable, least-captured data in the car industry** — the buyer's real pre-purchase intent and decision behaviour — running **privately on-prem** so it clears the privacy bar that kills everyone else, and **closing the loop from the showroom floor all the way to the OEM.**

The problem

The automotive purchase is one of the largest, most considered, most emotional consumer decisions there is — and the data about *why* buyers choose, hesitate, and walk is almost entirely **uncaptured**. Not under-captured: structurally uncapturable, because the only people positioned to capture it — salespeople — can't (why). The industry runs on memory, sticky notes, lagging sales figures, and survey-with-a-prize-draw guesswork.

Why now

Three curves crossed at the same time:



For the first time it is possible to **capture** this data (agentic AI), **do it privately** (local models on-prem), and **act on it** autonomously — all at once. That simultaneity is the window.

What it is

An autonomous agent across four capabilities — unify the dealership's customer data, model behaviour, generate custom insights, and turn them into decisions and actions — delivered through McDonald's-style kiosks and iPad nodes on the floor, and running entirely on the dealership's own hardware.

See the architecture and the raw-data-to-action scenario.

The wedge: the most valuable data in autos

Value = stakes × relationship length × decision-signal × scarcity × reach up the chain. This data scores at the top of all five (detail):

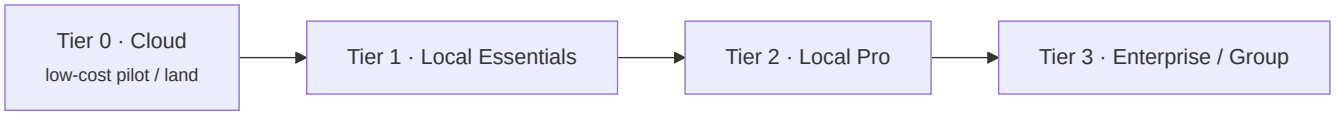
- **Huge stakes per decision** — a vehicle, a trade-in, finance, years of service.
- **A lifelong, repeating relationship** — shop → buy → own → service → trade → repeat.
- **Genuine decision signal**, not vanity metrics.
- **Scarce** — basically nobody has it, because it was uncapturable.
- **Valuable at every level** — salesperson, dealer, group, and **OEM**.

The moat

The AI is commodity. The moat is **capture + privacy + the loop**:

Layer	Why it defends
Data capture	Being in every channel at once, capturing as a by-product — hard to replicate without the on-floor footprint
On-prem/private	The only posture that can legally hold this PII at scale as law tightens (privacy)
The OEM loop	Once aggregated buyer-intent flows to manufacturers and shapes product/pricing, it compounds and entrenches (loop)
Data network effect	Every interaction sharpens the models; more dealers → more signal → better product

Business model: four tiers as a pricing ladder



- **Land** with low-cost Tier 0 pilots to prove value fast.
- **Expand** to on-prem Tiers 1–3 (hardware + license/subscription) once proven — the identical workflows migrate with no rebuild.
- **Multiply** across a dealer group (Tier 3) and, longer term, **monetize the aggregated OEM feedback loop** — a second revenue surface the data unlocks.

Revenue surfaces: setup/hardware, recurring license per site/seat, group/enterprise contracts, and OEM-facing insight products.

Market

Tens of thousands of dealerships; a global new+used vehicle market measured in the trillions; OEMs that spend heavily to *infer* buyer intent they could instead measure. The beachhead is a single rooftop; the expansion is the group; the prize is the loop to the manufacturer.

> **On numbers**
>
> Market and revenue framing here is directional for positioning. A bottoms-up model (per-rooftop economics, attach rates, hardware vs. recurring split) is a fast-follow once a design-partner dealer is signed.

Go-to-market

1. **Design-partner dealership** — Tier 0/1, prove conversion, trade-in capture, and retention lift against a holdout.
2. **Reference rooftop → group** — expand within the group on Tiers 2/3.
3. **OEM conversation** — bring aggregated, anonymized buyer-intent signal to the manufacturer; open the loop revenue.

Roadmap & the ask

Today: concept and documentation complete (roadmap). **Next (Phase 1):** specifications + a measured design-partner pilot.

The ask — three ways to help:

1. **Capital** — a `[pre-seed / seed — $XXXk–$X.Xm — to confirm]` round to fund Phase 1 specs and a Tier-0/1 pilot that proves the conversion, trade-in, and retention lift against a holdout.
2. **A design-partner dealership** — one rooftop willing to run the pilot and share outcomes; they get the lift and first-mover advantage.
3. **An OEM introduction** — a manufacturer contact to scope the aggregated, anonymized buyer-intent feedback loop.

Use of funds: specifications & architecture, the local-model evaluation and first models, integration with the pilot dealer's systems, and the kiosk/iPad build.

> **Set the numbers**
>
> The round, amount, and valuation are placeholders (`[...]`) until confirmed — tell me the real figures and I'll drop them in. The strategic asks (design partner, OEM intro) hold regardless.

> **Why we win**
>
> Everyone has an AI agent. We found the gold it's for — the car industry's most valuable data — and we're the only ones capturing it privately and closing the loop to the OEM. The technology is ordinary; the position is not.

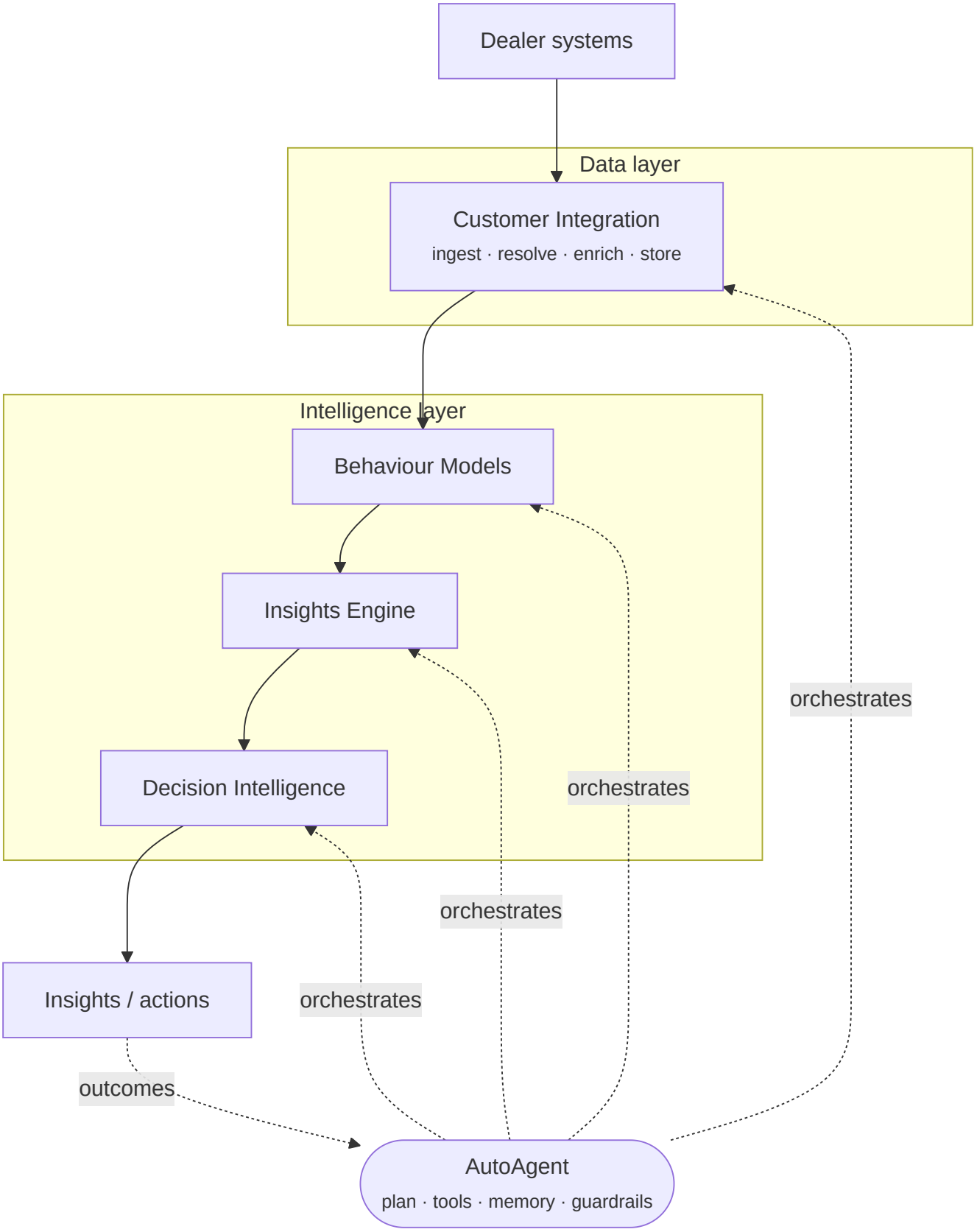
AutoAgentAI — Technical Overview

A dense, engineer-facing tour for someone evaluating *how it's built* and *whether it holds up*. For the full treatment, follow the links into the detailed docs.

> **System in one sentence**
>
> A stateful, autonomous agent orchestrates four capability pillars — data integration, behaviour modeling, insight generation, and decision intelligence — over a dealership's unified customer data, running on local open-source models, on-prem, behind thin kiosk/iPad clients.

Architecture

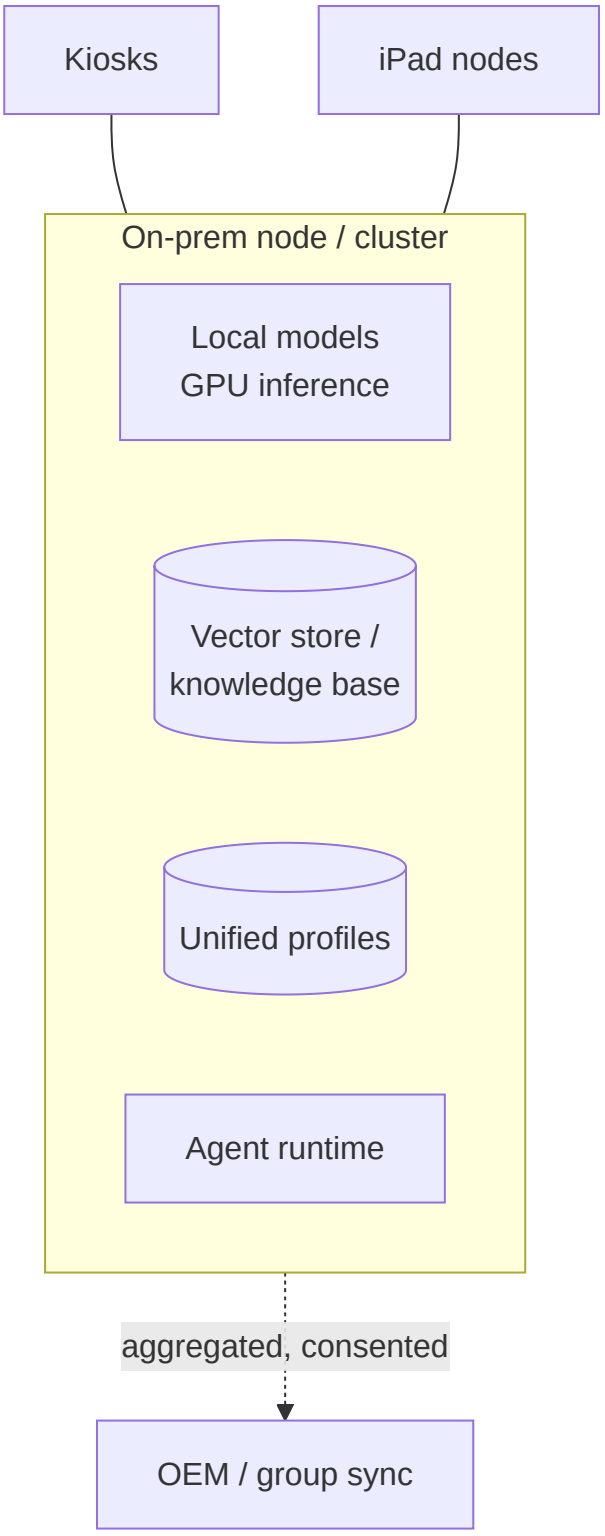
Four pillars in a pipeline, driven by a fifth (the agent). Full detail: Architecture.


Dealer systems
Data layer
Customer Integration
ingest · resolve · enrich · store
Intelligence layer
Behaviour Models
Insights Engine
Decision Intelligence
Insights / actions
outcomes
orchestrates
orchestrates
orchestrates
orchestrates
AutoAgent
plan · tools · memory · guardrails

Layer	Responsibility	Detail
Source	Dealer systems: DMS/CRM, web, service, finance	—
Data	Ingest, identity resolution, enrichment, unified profiles	Customer Integration
Intelligence	Propensities, segments, insights, next-best-action	Models · Insights · Decisions
Agent	Orchestration, memory, autonomy, audit	Agent Loop
Experience	Kiosks, iPad nodes, approvals, chat, API	Hardware

Deployment topology

Thin clients, intelligent core. Kiosks and iPads are **nodes**, not stores — models and data stay on the on-prem box.



Four tiers scale the same stack from a single workstation (Tier 1) to a redundant GPU cluster (Tier 3); Tier 0 is a cloud variant for non-sensitive pilots. See the tier matrix.

Local model strategy

Tasks are bounded — read, extract, classify, summarize, rank, draft — and **retrieval-grounded** + **structured-output-constrained**, which is precisely where open local models match cloud models. Rationale and limits: Local Models Are Enough.

- **Grounding:** RAG over the local knowledge base; the model reasons over supplied facts, not parametric memory.
- **Structure:** schema-constrained outputs (fill fields, pick from menus) over freeform generation.
- **Decomposition:** multi-step logic lives in the agent/tools, not one giant call; each inference is small and verifiable.
- **Economics:** fixed-cost inference enables a continuously-running loop without a per-call meter.

Data model & identity

Conceptual model (physical schema deferred to specs): a `Customer` with many `Identity`, `Event`, `Order`, `Interaction`, and `Feature` records. Identity resolution is deterministic-first (strong keys), probabilistic with confidence scoring for the rest, and **confidence-aware** (low-confidence links are flagged, not silently merged). Detail: Customer Integration.

Agent runtime

Perceive → reason/plan → act → learn, on its own cadence (scheduled + event-driven). Three memory types (working, episodic, semantic). Tools = the four pillars + dealer systems. Detail: Agent Loop.

Security, privacy & guardrails

- **Data locality** is the primary control: on-prem means no third-party processors, no cross-border transfer, minimal breach surface (privacy).
- **Guardrails:** graduated autonomy L0–L4; confidence floors; budget/frequency caps; excluded actions; escalation.
- **Provenance:** every insight/action carries a trace (data → reasoning → output) for explainability and audit of automated decisions.
- **Safety:** reversible actions, kill-switch, external text treated as data not instructions.
- **Thin clients:** a lost iPad is a window, not a database copy.

What's decided vs. deferred

Decided (logical)	Deferred to Phase-1 specs (physical)
Five-pillar architecture; on-prem-first; thin clients	Concrete stack, languages, datastores
Local-model-first; retrieval-grounded; structured outputs	Specific model(s), GPU sizing, eval bar
Graduated autonomy + provenance	Default guardrail envelopes, action catalog v1
Conceptual data model	Physical schema, ingestion (stream vs batch)

Deliberately logical-first so the concept isn't prematurely locked to a stack; ADRs capture load-bearing choices when specs begin. See Specifications and the Roadmap.

> **Evaluation is a first-class Phase-1 deliverable**
>
> A task-level eval suite on real dealership data proves the chosen local model clears the quality bar before anything ships — the claim in Local Models Are Enough is to be *measured*, not asserted.

AutoAgentAI — For Sales Managers

Straight talk, no tech jargon. Here's what this thing actually does for your store.

> **In plain English**
>
> It's a smart assistant that quietly watches every customer — online, on the lot, in service — remembers *everything*, and tells your people exactly who to call, what they care about, and how to help them buy. It never forgets a lead, and it runs on a box in your back office, so your customer info never leaves the building.

The problem you already know

- Leads fall through the cracks. Somebody gets busy, forgets to follow up, and that deal walks to the store down the road.
- Your best closers keep it all in their head. When they leave, it walks out with them.
- Half the real story never gets written down — the customer who loved the SUV but worried about gas money, the trade-in that's about to die. Your people can't sell *and* type notes at the same time, so they (rightly) stay present and the details vanish.
- You find out *why* you lost a deal… never. You just see the number at month-end.

None of that is your people being lazy. It's impossible for a human to track every customer across the website, the showroom, the phone, and service all at once. A computer can. That's the whole idea.

What it does for your store

What you want	How it helps
Stop losing leads	It tracks every customer automatically and tells your rep who to follow up with, today, and why.
Know your customer before you walk up	"This is Maria — second kid coming, loved the hybrid, worried about running costs, her current car's on its last legs." On your rep's iPad, before hello.
Sell more, without pressure	Your rep leads with the answer to the customer's real worry instead of a generic pitch. People buy when they feel understood.
Catch more trade-ins	It spots when someone's car is near the end and flags it.
Keep service customers coming back	It notices who's drifting and nudges before they're gone.
Never lose the knowledge	When a rep leaves, everything they knew about every customer stays in the system.

What your salespeople's day looks like

- They carry an **iPad**. It's their window into the whole system — anywhere on the lot.
- Customers can use **self-serve kiosks** (like the ones at McDonald's) — easy, no-pressure, and they quietly tell the system what the customer's interested in.
- Instead of guessing who to chase, your rep opens the iPad and sees a short list: *these* customers, *this* is what they need, *here's* what to say.
- The boring admin — logging notes, remembering follow-ups — the system does it.

"Is this going to replace my salespeople?"

Honest answer: it changes the job more than it replaces people.

- The parts that go away are the parts nobody liked anyway — chasing, guessing, and data entry.
- The part that stays and matters more is the **human** part: making a nervous buyer feel taken care of. The computer can't do that. Your people can.

- You'll likely run a **leaner, sharper team** where each person handles more customers and gives each one *more* attention, because the busywork is gone. Your closers become trusted advisors, not pressure salesmen.

Think of it as giving every salesperson a tireless assistant who remembers everything and does the paperwork.

Your info stays yours

This runs on a computer **in your dealership**. Your customers' names, numbers, and details don't get shipped off to some cloud company. That keeps you on the right side of privacy laws and lets you tell customers, honestly, "your information stays here with us." (More on that if you care: the privacy page.)

You can start cheap and simple to try it out, then move to the in-house setup once you see it working. Four options, from a low-cost trial to a full in-store system — the four setups.

What you have to do

Not much. The system does the watching, remembering, and suggesting. Your job stays the same: lead a team that takes great care of customers — just with way better information and nothing falling through the cracks.

> **The bottom line**
>
> Every customer tracked. No lead lost. Your reps walk up already knowing what matters. More sales, more trade-ins, more repeat service — and your customer info never leaves the building.

Want to see how one customer goes from "just looking" to "sold"? Read the quick story: From Raw Data to a Decision.

Vision

The title, decoded

The project's full name is a compact specification. Read left to right:

> **AutoAgent** · and **Customer-Integrated** · **Custom Insights** · into **Purchase** · and **Overall Behaviour** · and **Decision Making**.

- **AutoAgent** — the system is an *autonomous agent*. It has goals, it acts on its own cadence, and it decides what question to ask next instead of waiting to be queried.
- **Customer-Integrated** — it is wired into the business's customer data and systems. Insights are grounded in *this* business's customers, not a generic model.
- **Custom Insights** — output is *tailored*: specific to the business, the segment, and often the individual customer.
- **into Purchase** — a first-class focus on buying behaviour: conversion, basket, frequency, value, churn.
- **and Overall Behaviour** — the whole customer relationship: engagement, usage, support interactions, sentiment, lifecycle stage.
- **and Decision Making** — insights exist to drive *decisions* — the agent frames findings as actions a human or system can take, and can take some itself.

Vision statement

> **Every business should have an analyst that never sleeps, never forgets a customer, and always ties what it finds to a decision.**
> AutoAgentAI is that analyst as an autonomous agent — it reads all the customer signal a business generates, learns how its customers buy and behave, and continuously turns that into decisions that grow the relationship.

What success looks like

=== "For the business"

```
- A single, trustworthy, resolved view of each customer.
- A steady stream of *decisions*, not just charts: "offer X to segment Y,"
  "these 300 accounts are about to churn — here's why and what to do."
- Measurably better outcomes: higher conversion, retention, and lifetime value,
  with the agent's contribution attributable.
```

=== "For the operator (analyst / marketer / ops)"

```
- Hours of manual slicing-and-dicing replaced by reviewing and approving
  agent-proposed actions.
- Ability to ask the agent questions in plain language and get grounded answers.
- Trust: every insight traces back to the data and reasoning behind it.
```

=== "For the customer (end result)"

```
- More relevant offers and less noise.
- Problems anticipated before they become complaints.
- A relationship that feels understood rather than mass-marketed to.
```

Guiding principles

1. **Decision-first.** An insight that doesn't change a decision is noise. Every output is framed as "so what should we do?"
2. **Grounded and explainable.** Every insight traces to the data and reasoning that produced it. No black-box pronouncements.
3. **Autonomous but accountable.** The agent acts on its own cadence, but within guardrails, with human-in-the-loop where stakes are high, and with a full audit trail.
4. **Custom over generic.** Models and insights adapt to each business; the product is not a one-size-fits-all benchmark report.

5. **Privacy and trust by design.** Customer data is sensitive; consent, minimization, and security are foundations, not afterthoughts.

Non-goals (for now)

- Not a general-purpose BI tool — it is opinionated toward customer behaviour and decisions.
- Not a data warehouse — it integrates with and reads from existing stores rather than replacing them.
- Not a fully hands-off autopilot on day one — autonomy is earned progressively as trust and guardrails mature (see the roadmap).

> **This framing is a draft**
>
> The decoding above is the current working interpretation of the project name. If the intended meaning differs, this page is the right place to correct it — everything else in the docs follows from here.

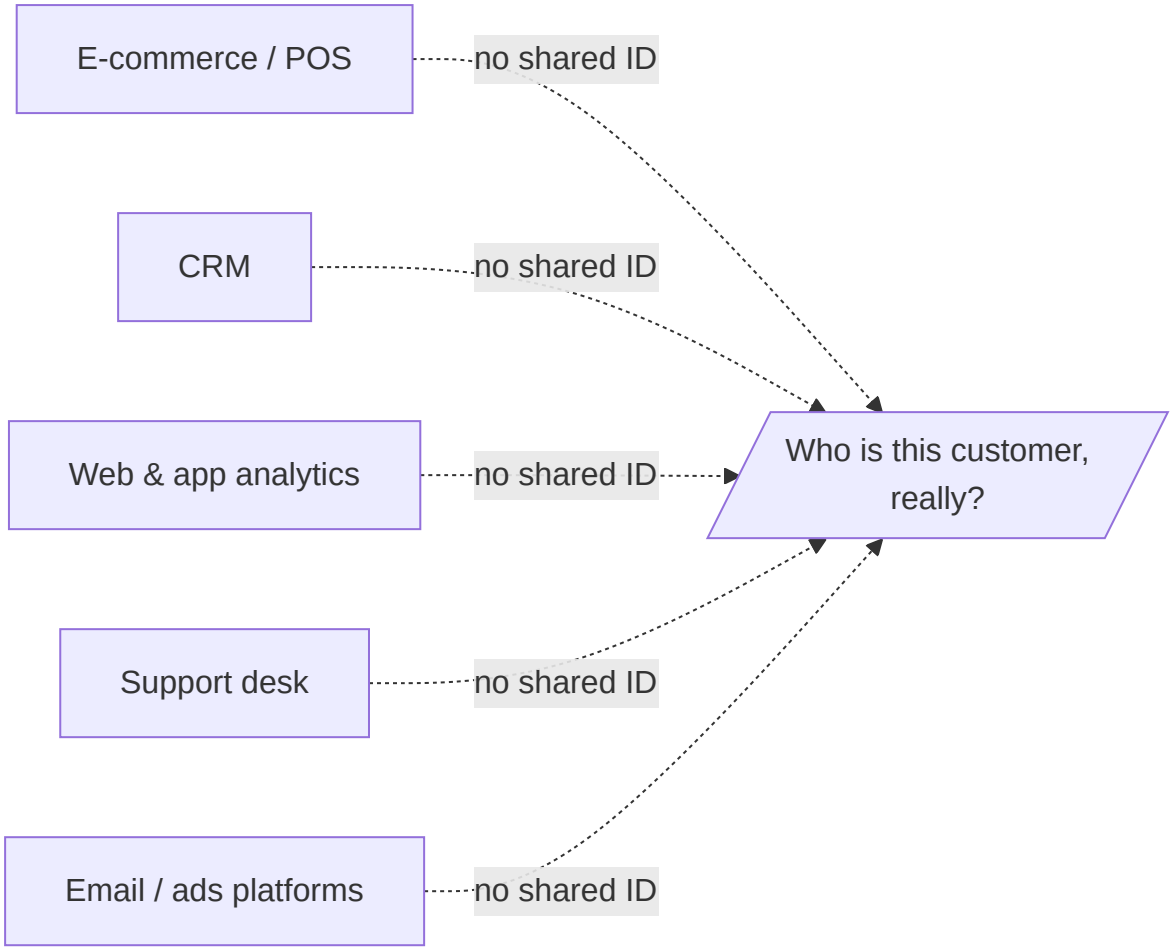
Problem & Motivation

The problem

Businesses generate more customer signal than ever, and act on less of it than they think. Three gaps recur:

1. Data is fragmented

Customer information lives in separate systems that rarely agree on who a customer is or what they did:



The same person is a row in the POS, a contact in the CRM, an anonymous device in analytics, and a ticket in support — with nothing tying them together.

2. Insight is reactive and manual

Even where data is joined, the standard tool is a **dashboard**: it shows metrics and waits for a human to notice something, form a hypothesis, and go dig. The questions asked are limited by the analyst's time and imagination, and most patterns are never looked for.

3. Insight rarely becomes a decision

The last mile is the hardest. A chart showing churn rising does not, by itself, tell anyone *which* customers, *why*, or *what to do*. The translation from "interesting finding" to "specific action taken" is where most value leaks out.

Why now

- **Agentic AI is viable.** LLM-based agents can plan, use tools, query data, and explain their reasoning — making an "analyst that works on its own" realistic rather than aspirational.

- **Customer data is more connectable.** Modern data stacks (warehouses, CDPs, event pipelines) make integration tractable where it used to be bespoke.
- **Expectations have shifted.** Customers expect relevance; businesses expect their tools to be proactive. Static reporting no longer meets either.

Who it's for

Audience	Pain today	What AutoAgentAI gives them
Growth / marketing teams	Segments go stale; targeting is coarse	Live, behaviour-based segments and next-best-action per customer
Product teams	Hard to see how usage predicts retention or expansion	Behaviour models linking usage to outcomes
Customer success / retention	Churn noticed too late	Early, explained churn risk with recommended saves
Analysts / data teams	Flooded with ad-hoc requests	An agent that handles the routine slicing and surfaces the non-obvious
Leadership	Decisions made on gut or lagging reports	Decision-ready insight tied to measurable outcomes

The core bet

> If an autonomous agent **integrates** the customer data, **continuously** generates insight tailored to the business, and frames every insight as a **decision**, then businesses act on far more of their signal — and the gap between "data we have" and "decisions we make" closes.

This is the thesis the rest of the documentation builds on. The architecture is the structure designed to deliver it; the use cases make it concrete.

Concepts & Glossary

A shared vocabulary so the rest of the docs — and later the specs — mean the same thing by the same words.

Core concepts

AutoAgent

The autonomous orchestrator at the heart of the system. It holds goals, decides what to investigate, invokes the other components as tools, and chooses what to surface or act on. Distinguished from a passive report by **initiative** (it acts without being asked) and **continuity** (it runs on its own cadence). See The AutoAgent Loop.

Customer-Integrated

The property of being wired into a business's actual customer data and systems. Concretely: ingesting from the business's sources, resolving identities into one view per customer, and being able to read context and (eventually) trigger actions in those systems. See Customer Integration.

Custom Insight

A finding that is **tailored** (specific to this business / segment / customer), **grounded** (traceable to the data and reasoning behind it), and **decision-oriented** (it implies an action). Contrast with a generic benchmark or a raw metric. See Insights Engine.

Purchase behaviour

Everything about how a customer buys: what, when, how often, how much, through which channel, and in response to what. Includes conversion, basket composition, recency/frequency/monetary (RFM) patterns, price sensitivity, and churn.

Overall behaviour

The full relationship beyond transactions: engagement, product/feature usage, support interactions, content consumption, sentiment, and lifecycle stage. Purchase behaviour is a subset of overall behaviour.

Decision Making

The discipline of turning insight into action. In AutoAgentAI every insight is framed as a decision — a recommended next-best-action, a targeting choice, an intervention — with expected impact and a way to measure the result. See Decision Making.

The five pillars

#	Pillar	Role
1	**Customer Integration Layer**	Connect, unify, resolve, and enrich customer data
2	**Insights Engine**	Generate custom, grounded, decision-oriented insights
3	**Behaviour Models**	Model purchase and overall behaviour (segments, propensities, journeys)
4	**Decision Intelligence**	Turn insights into recommendations and actions
5	**The AutoAgent**	Orchestrate the loop; hold goals, guardrails, and memory

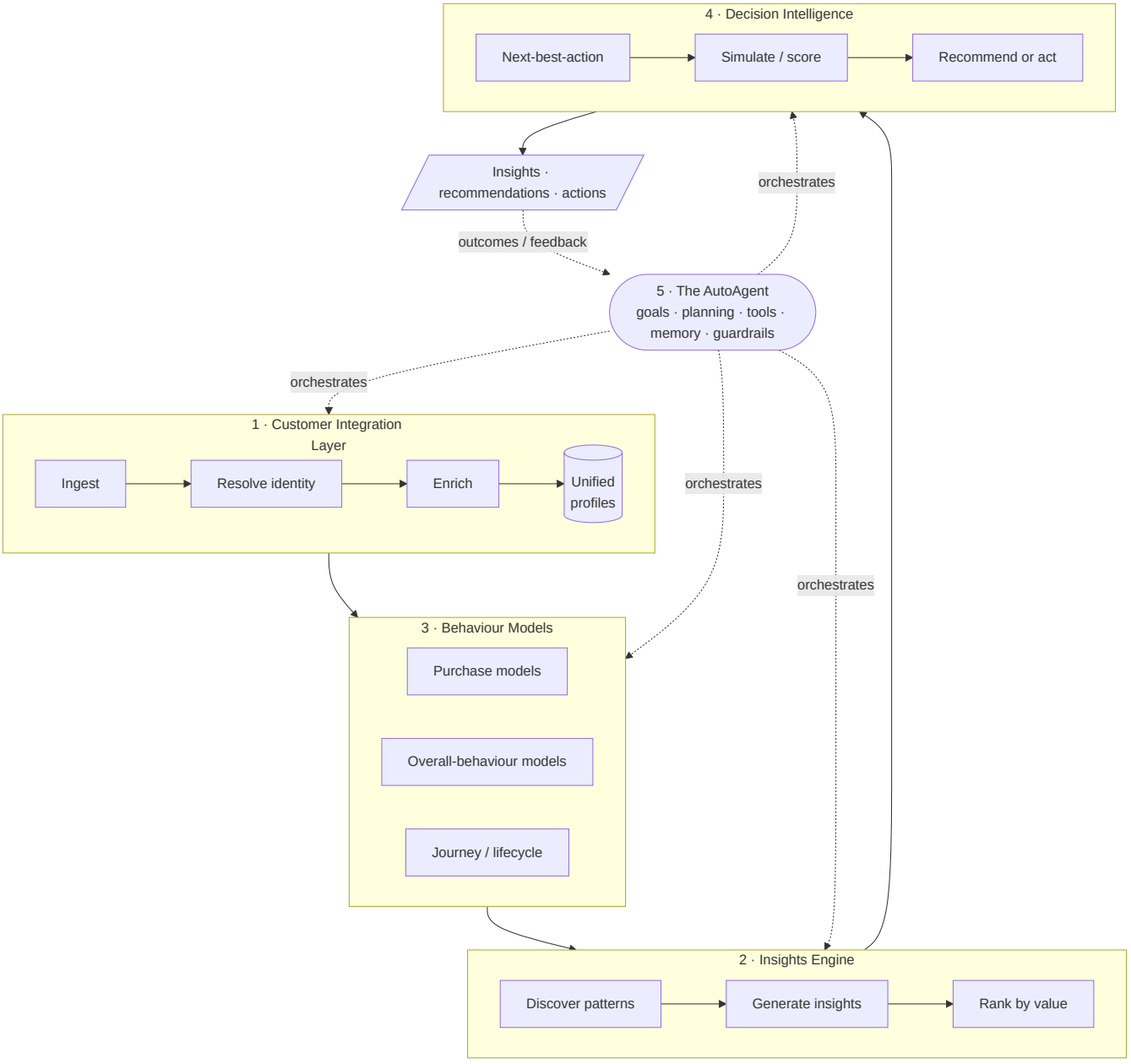
Glossary

Term	Definition
Action	A concrete step the agent recommends or takes (send offer, flag account, adjust segment).
Attribution	Measuring the outcome and impact caused by an action the agent took.
CDP	Customer Data Platform — a system that unifies customer data; AutoAgentAI can sit on one or provide its own integration layer.
Churn	A customer ending or lapsing their relationship (stopping purchases, cancelling).
Cohort	A group of customers sharing a defining trait or start time, analyzed together.
Customer 360 / Unified profile	One resolved record per customer combining all sources.
Entity / Identity resolution	Deciding that records across sources refer to the same customer.
Feature (model sense)	A computed signal about a customer used as model input (e.g. days-since-last-order).
Guardrail	A rule constraining what the agent may do autonomously (budget caps, approval gates, excluded actions).
Human-in-the-loop (HITL)	A person reviews/approves agent proposals before they take effect.
Insight	See *Custom Insight*.
LTV / CLV	(Customer) Lifetime Value — total value a customer is expected to generate.
Next-best-action (NBA)	The single most valuable action to take for a given customer right now.
Propensity	A modeled probability a customer will do something (buy, churn, upgrade).
RFM	Recency, Frequency, Monetary — a classic purchase-behaviour segmentation.
Segment	A defined group of customers sharing behaviour or attributes.
Signal	Any piece of customer data that may carry meaning.
Tool (agent sense)	A capability the agent can invoke — a query, a model, an API call.
Trace / Provenance	The recorded chain from raw data → reasoning → insight → action, enabling explanation and audit.

> **Naming discipline**
>
> When specs are written, terms defined here should be used verbatim. If a spec needs a new term, add it here first.

Architecture

AutoAgentAI is organized into **five pillars**. Four are capabilities arranged as a pipeline; the fifth — the AutoAgent — is the orchestrator that drives them, holds goals and guardrails, and closes the loop with feedback.

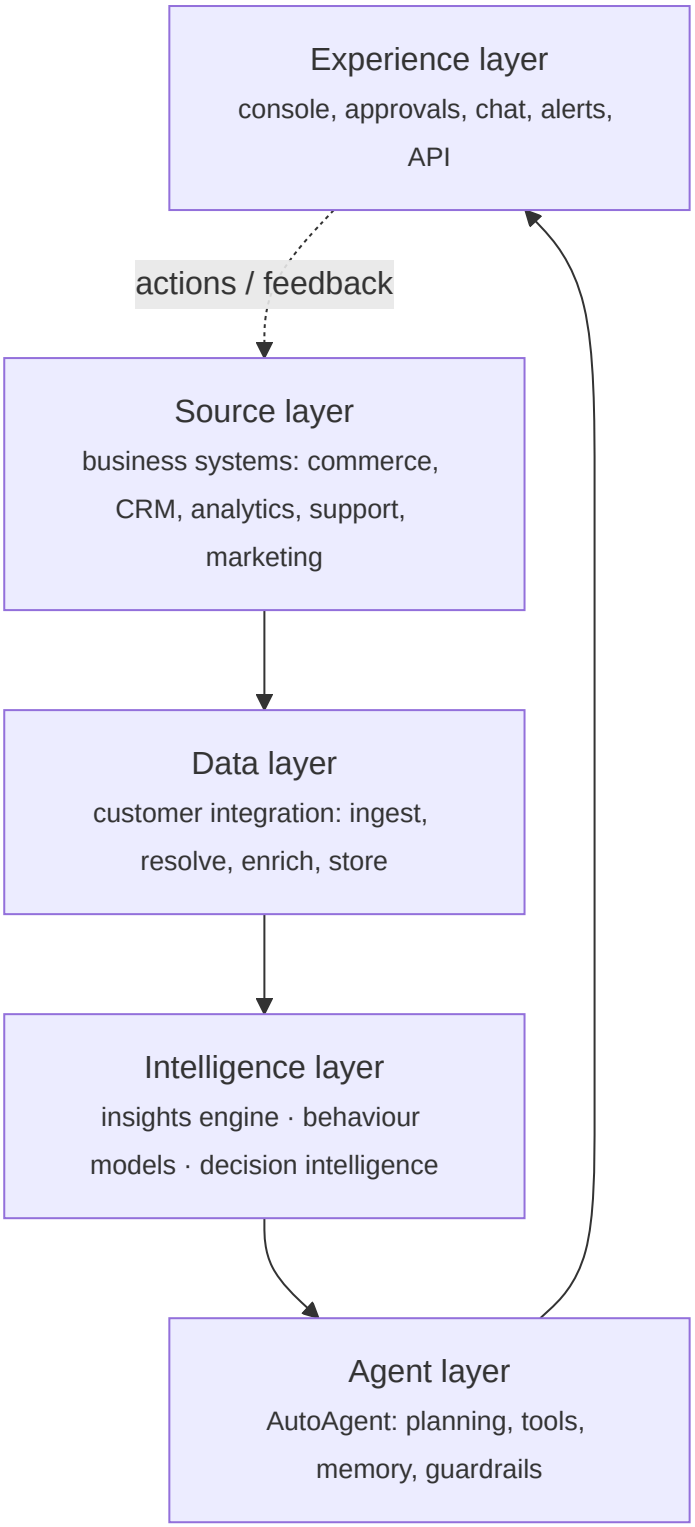


The pillars at a glance

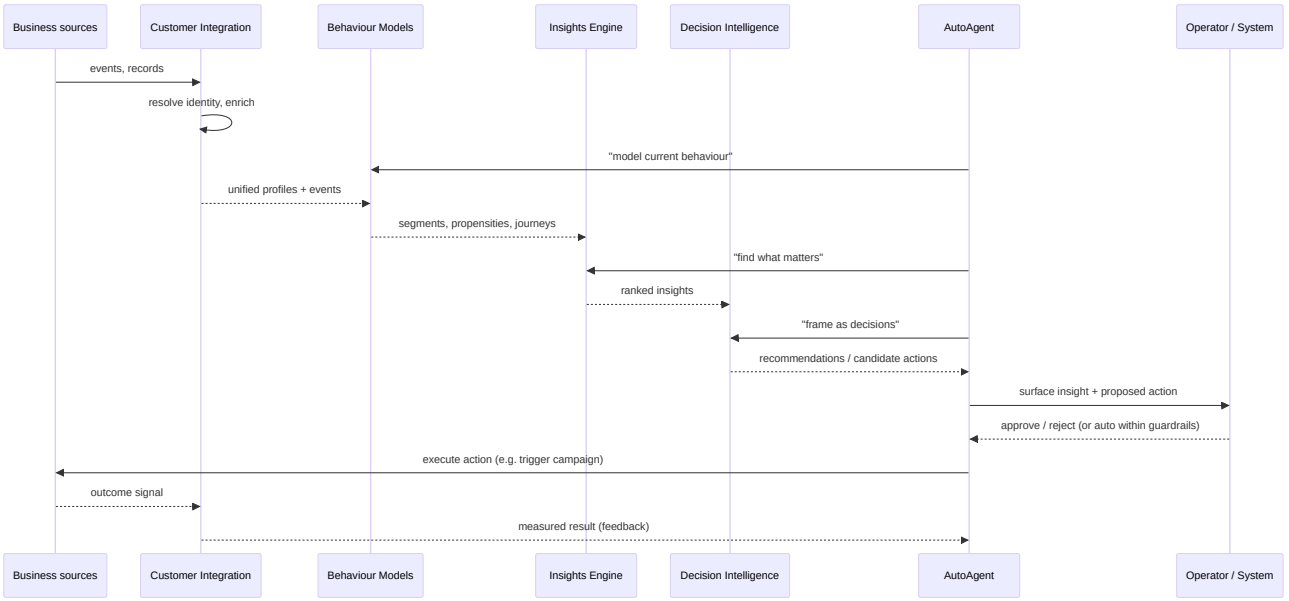
Pillar	Input	Output	Detailed page
1 · Customer Integration Layer	Raw data from business sources	Unified, enriched customer profiles + event history	Customer Integration
2 · Insights Engine	Profiles + model outputs	Ranked, grounded custom insights	Insights Engine
3 · Behaviour Models	Unified profiles + events	Segments, propensities, journeys	Behaviour Models
4 · Decision Intelligence	Insights + model outputs	Recommendations and actions	Decision Making
5 · The AutoAgent	Goals + everything above	Orchestration, autonomy, memory, audit	The AutoAgent Loop

Logical layers

A different cut of the same system — by layer rather than by pillar — useful when thinking about where a given responsibility lives:


Experience layer
console, approvals, chat, alerts, API
actions / feedback
Source layer
business systems: commerce, CRM, analytics, support, marketing
Data layer
customer integration: ingest, resolve, enrich, store
Intelligence layer
insights engine · behaviour models · decision intelligence
Agent layer
AutoAgent: planning, tools, memory, guardrails

Data flow (end to end)



Cross-cutting concerns

These apply to every pillar and must be designed in, not bolted on:

- **Trust & provenance** — every insight and action carries a trace from raw data through reasoning to output, enabling explanation and audit.
- **Privacy & consent** — identity resolution and action-taking respect consent, data-minimization, and regional rules (e.g. GDPR/CCPA). Covered further in Customer Integration.
- **Guardrails & autonomy level** — what the agent may do without approval is explicit and configurable. See The AutoAgent Loop.
- **Observability** — the system's own behaviour (what it looked at, decided, and did) is logged and measurable.
- **Extensibility** — new data sources, models, and action types plug in without reworking the core.

Design stance (intentionally not yet decided)

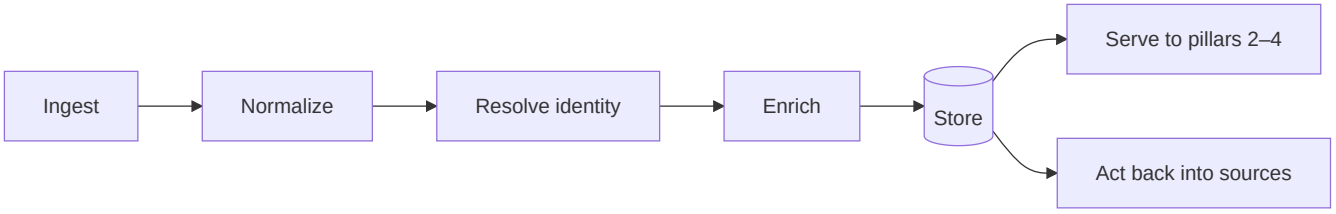
This document describes the **logical** architecture. The **physical** choices — languages, model providers, datastores, deployment — are deliberately deferred to the specifications so the concept isn't prematurely locked to a stack. Where a choice is load-bearing, it will be recorded as a decision with trade-offs when specs begin.

Customer Integration Layer

Pillar 1. The foundation: turn fragmented, multi-source customer data into one trustworthy, resolved, enriched view per customer — and provide the read/write connection back to the business's systems.

> Nothing downstream is better than this layer. Insights and decisions inherit the quality, coverage, and trustworthiness of the integration beneath them.

Responsibilities



1. **Ingest** — pull/receive data from the business's sources.
2. **Normalize** — map disparate schemas into a common model (customers, events, products, interactions).
3. **Resolve identity** — decide which records across sources are the same person.
4. **Enrich** — derive attributes and features (RFM, tenure, engagement scores).
5. **Store & serve** — maintain unified profiles + event history for the other pillars.
6. **Act** — provide the outbound connection so the agent can trigger actions (send a campaign, create a task, update a segment).

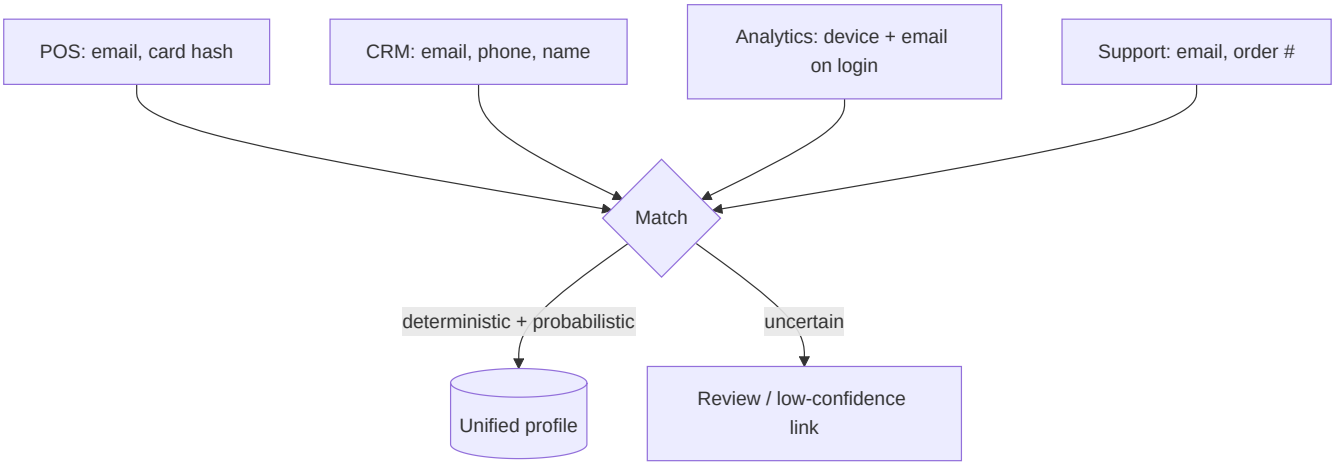
Source taxonomy

Category	Examples	Typical signal
Commerce / transactions	Shopify, POS, billing, subscriptions	What/when/how much bought
CRM	Salesforce, HubSpot	Accounts, contacts, pipeline, notes
Product / behavioural	Web & app analytics, event streams	Sessions, feature usage, clicks
Support	Zendesk, Intercom, tickets	Issues, resolution, satisfaction
Marketing	Email, ads, campaign platforms	Sends, opens, responses, spend
Voice of customer	Reviews, surveys, NPS	Sentiment, verbatim feedback

> **Build vs. sit-on-top**
>
> AutoAgentAI can either provide this layer itself or sit on an existing CDP / data warehouse and consume its unified profiles. The architecture supports both; which to lead with is a go-to-market and spec decision (see roadmap).

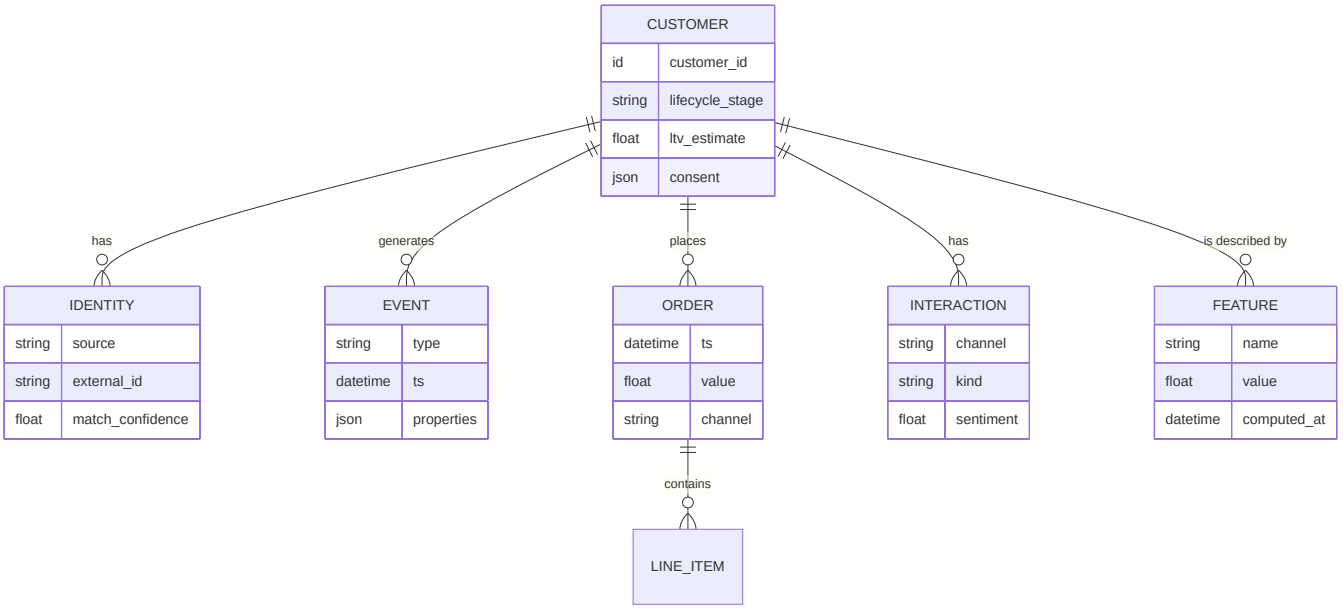
Identity resolution

The hardest and most valuable step: joining records that don't share a key.



- **Deterministic** matching on shared strong identifiers (email, phone, customer ID) first.
- **Probabilistic** matching on weaker combinations (name + address, device + behaviour) with a confidence score.
- **Confidence-aware**: low-confidence links are flagged rather than silently merged, because a bad merge corrupts every downstream insight.

The unified profile (conceptual model)



This is a **conceptual** model, not a schema — it shows the entities the rest of the system reasons about. Physical schema belongs in the specs.

Privacy, consent, and trust

Customer data is sensitive and regulated. This layer is where those obligations are enforced:

- **Consent-aware**: each profile records what the customer consented to; the agent may only use and act on data within that consent.
- **Data minimization**: collect and retain only what's needed for the decisions being made.
- **Regional rules**: support deletion/export requests and regional constraints (GDPR, CCPA, and similar).
- **Security**: encryption in transit and at rest; least-privilege access; auditable access logs.
- **Provenance**: every derived attribute records how and when it was computed, so downstream insights can be explained.

Open questions (for specs)

- Real-time streaming vs. batch ingestion — and which sources need which.
- Where unified profiles are materialized (own store vs. warehouse vs. CDP).
- The exact match/merge policy and confidence thresholds.

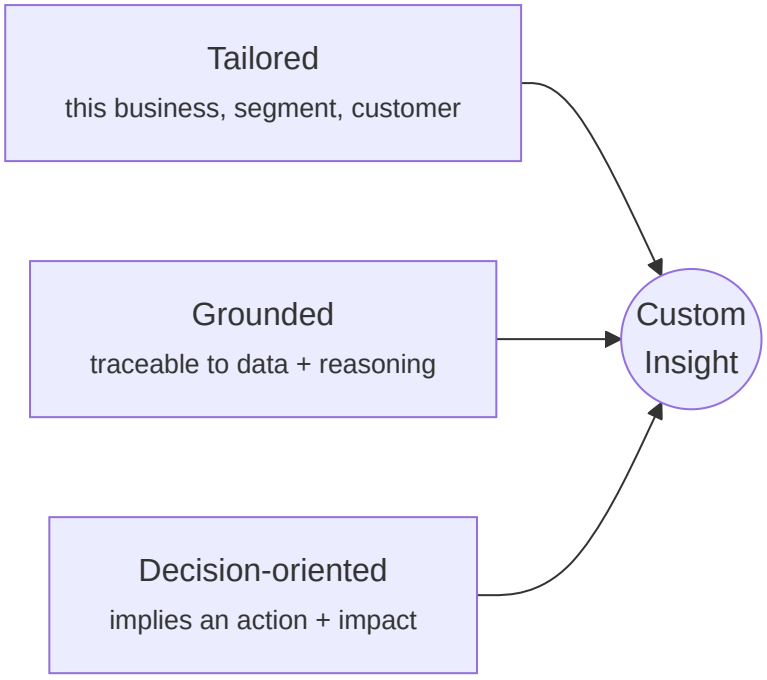
- How outbound actions are authorized and rate-limited per source.

Insights Engine

Pillar 2. Turns unified data and model outputs into **custom insights** — findings that are tailored, grounded, and decision-oriented.

> An insight here is not a metric and not a chart. It is a *claim about the business's customers, backed by evidence, that implies an action.*

What makes an insight "custom"



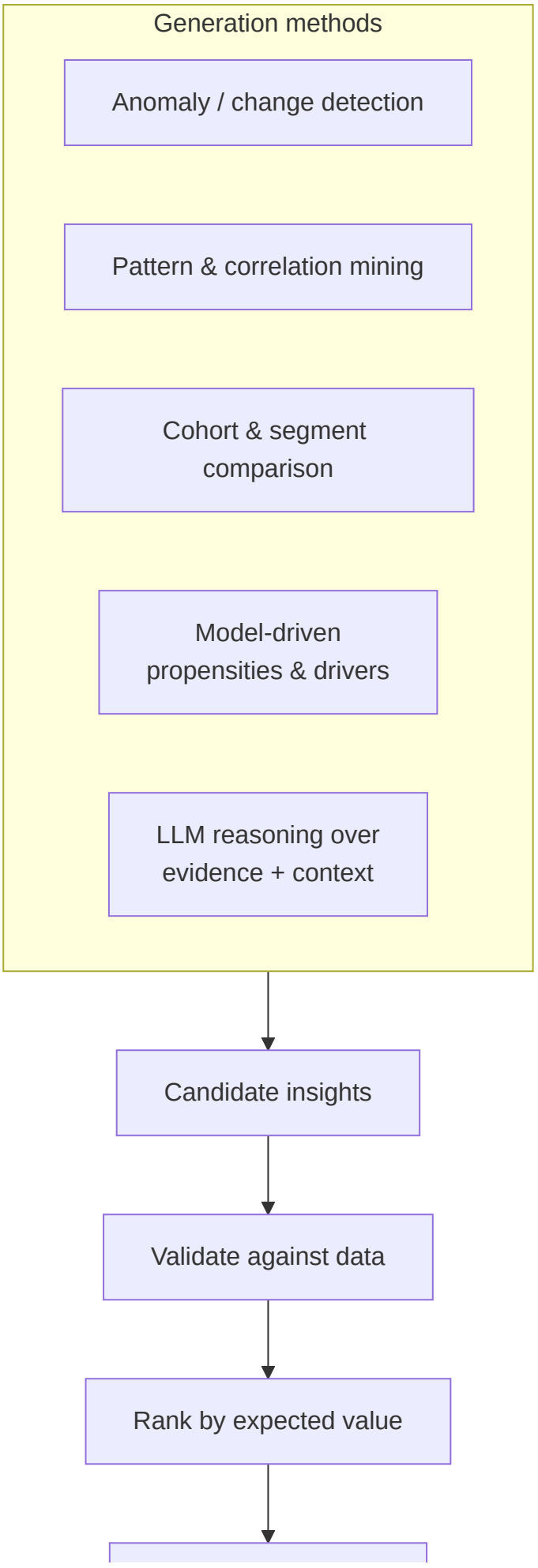
A finding earns the name only when all three hold. "Average order value is $48" is a metric. "Customers who buy accessories within 14 days of their first order have 2.3× the 12-month LTV, and 60% of first-time buyers never see an accessory offer" is an insight — tailored, grounded, and pointing at an action.

Anatomy of an insight

Field	Meaning
Statement	The claim, in plain language.
Evidence	The data, segment, and model output behind it.
Confidence	How strongly the data supports it.
Scope	Which customers/segment it applies to and how many.
Expected impact	What acting on it is worth (e.g. estimated revenue/retention).
Suggested decision	The action it implies — handed to Decision Intelligence.
Provenance	The trace from raw data → reasoning → this statement.

How insights are generated

The engine combines complementary methods rather than relying on one:


Generation methods
Anomaly / change detection
Pattern & correlation mining
Cohort & segment comparison
Model-driven propensities & drivers
LLM reasoning over evidence + context
Candidate insights
Validate against data
Rank by expected value

Surface top insights

- **Anomaly / change detection** — what moved, where, and for whom.
- **Pattern & correlation mining** — recurring associations in behaviour.
- **Cohort & segment comparison** — who differs from whom and why it matters.
- **Model-driven** — propensities and their drivers from the behaviour models.
- **LLM reasoning** — synthesizes evidence and business context into a plain-language, decision-framed statement; also proposes *what to investigate next* for the agent.

> **Guard against spurious findings**
>
> More methods means more candidate claims, and more chances for false positives. Every candidate is **validated against the data** (effect size, significance, holdout checks) before it is surfaced, and confidence is reported honestly.

Ranking: value over volume

The engine's job is not to produce many insights but the *right* ones. Candidates are ranked by **expected value of acting**, roughly:

> **priority ≈ (impact if true) × (confidence) × (actionability) ÷ (effort/cost)**

Low-confidence, low-impact, or un-actionable findings are suppressed. The agent can tune this ranking toward current goals (e.g. prioritize retention this quarter).

Types of insight (illustrative)

Type	Example
Opportunity	"Segment X is under-offered product Y they're likely to want."
Risk	"These accounts show the early churn signature seen in past lapsers."
Driver	"Support response time is the strongest predictor of repeat purchase."
Segment shift	"A fast-growing new behavioural segment has emerged this month."
Anomaly	"Conversion for channel Z dropped 18% for mobile users only."
Lifecycle	"Customers stall at onboarding step 3; those who pass it retain 2×."

Interfaces

- **Pull**: an operator asks a question in plain language; the engine answers with a grounded insight.
- **Push**: the agent runs the engine continuously and surfaces high-value insights without being asked.

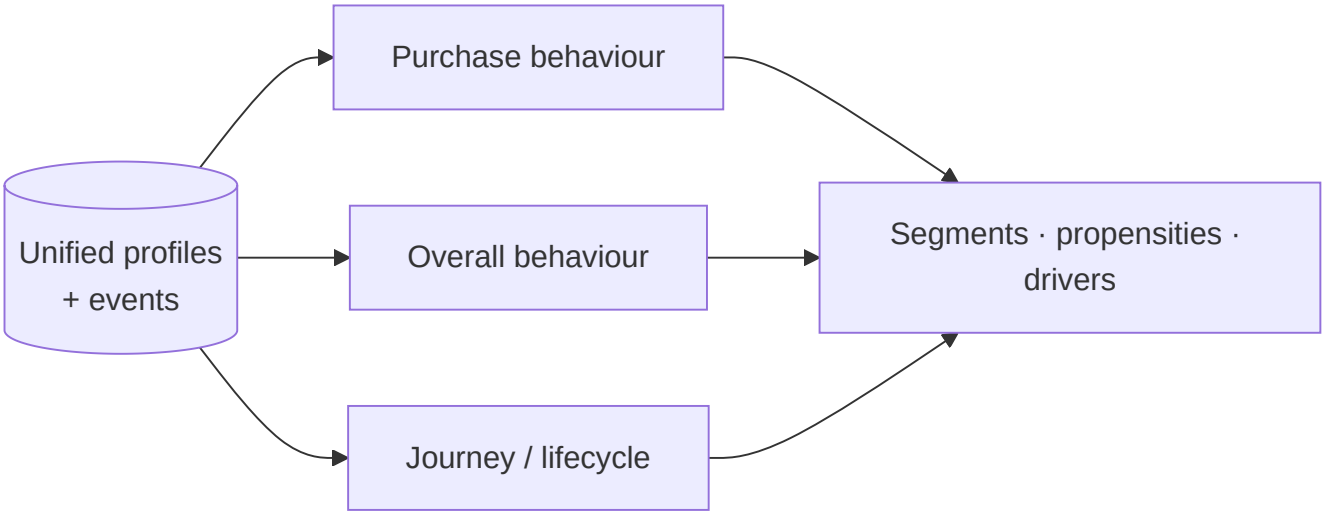
Both paths produce the same structured insight object, so everything is explainable and feeds Decision Intelligence identically.

Open questions (for specs)

- The exact validation/statistical bar before surfacing.
- How business context and goals are encoded for the LLM reasoning step.
- De-duplication and "insight fatigue" controls (not re-surfacing known findings).
- How confidence is calibrated and communicated.

Behaviour Models

Pillar 3. The quantitative understanding of customers: how they **purchase**, how they **behave overall**, and where they are in their **journey**. These models feed the Insights Engine and Decision Intelligence.



Purchase behaviour

How customers buy — the commercial core.

Model / measure	Question it answers
RFM (Recency, Frequency, Monetary)	How recently, often, and valuably does this customer buy?
Purchase propensity	How likely is this customer to buy (soon / this product)?
Basket / affinity	What products are bought together or in sequence?
Price sensitivity	How does this customer respond to price and promotions?
LTV / CLV	What is this customer worth over their lifetime?
Churn / lapse risk	How likely is this customer to stop buying?

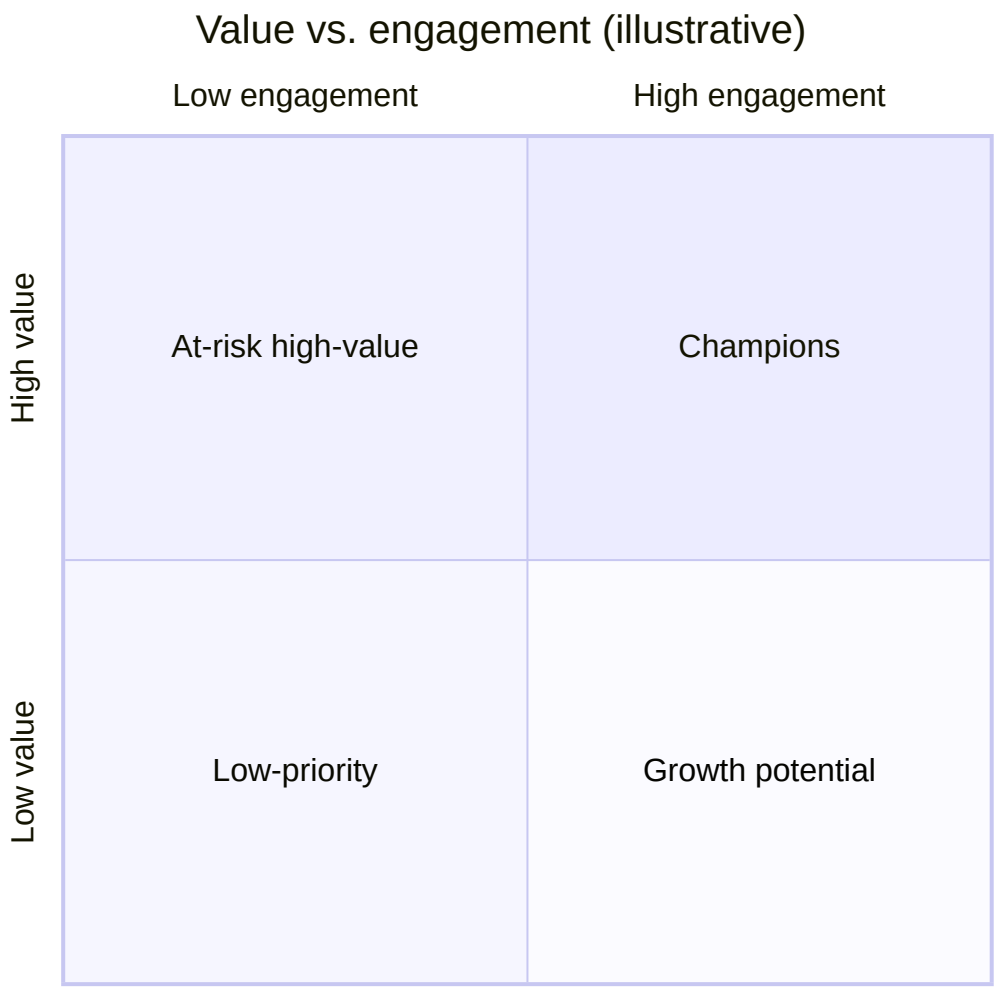
Overall behaviour

The relationship beyond the transaction — often the leading indicator of purchase behaviour.

Model / measure	Question it answers
Engagement score	How actively is the customer interacting (visits, usage, opens)?
Feature / product usage	What are they using, how deeply, and is it growing?
Support & satisfaction	How much friction are they hitting; how do they feel?
Sentiment	What is the tone of their feedback and interactions?
Channel preference	Where and how do they prefer to engage?

Behavioural segmentation

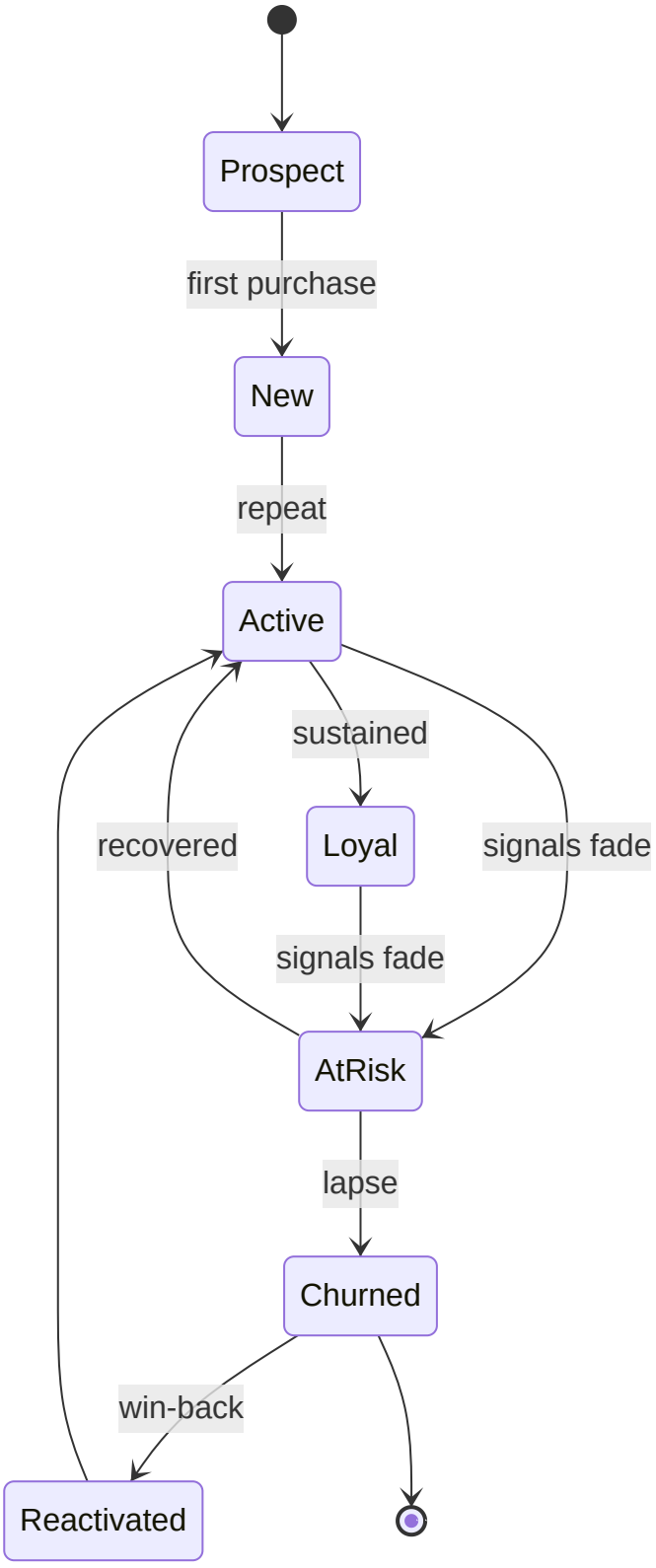
Rather than fixed demographic buckets, customers are grouped by *how they behave*, and segments are re-learned as behaviour shifts.



Segments are **living**: a customer can move between them, and the emergence of a new segment is itself an insight.

Journey & lifecycle

Where a customer is in their arc, and what typically comes next.



Transitions carry probabilities and drivers, so the agent knows not just *where* a customer is but *where they're heading* and *why*.

Modeling principles

1. **Explainable first.** Prefer models whose drivers can be surfaced; the Insights Engine needs the "why," not just the score.
2. **Features are shared.** Behavioural features computed here are reused across models and insights, defined once in the integration layer.
3. **Honest uncertainty.** Every prediction carries a confidence/interval; the agent treats a shaky prediction differently from a firm one.
4. **Monitored for drift.** Behaviour changes; models are watched for decay and retrained, and drift is itself a signal.

5. **Start simple.** Well-understood methods (RFM, logistic/propensity models, clustering) before complex ones — complexity is added only where it pays.

Open questions (for specs)

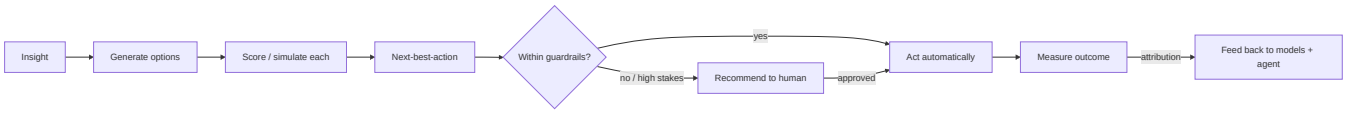
- Which models are v1 (likely RFM, churn/purchase propensity, behavioural clustering) vs. later.
- Build vs. buy for modeling infrastructure.
- Training/serving cadence (batch scores vs. on-demand).
- How model outputs are versioned so insights remain reproducible.

Decision Intelligence

Pillar 4. The last mile: turn insights and model outputs into **decisions** — specific recommendations and actions — and measure what they achieve.

> This pillar is why the product exists. An insight that never changes a decision delivered no value. Here, every insight becomes a *decision with an owner, an expected impact, and a way to tell whether it worked.*

From insight to decision



Next-best-action (NBA)

For a given customer or segment, the system determines the single most valuable action to take *now*, given goals, constraints, and predicted response.

Input	Role
Insight / propensity	What's true and likely about the customer
Action catalog	The menu of possible actions (offer, message, flag, nurture…)
Expected response	Modeled effect of each action on this customer
Goals & constraints	What the business is optimizing and its limits (budget, frequency caps)
Guardrails	What's allowed autonomously vs. needs approval

The output is a ranked recommendation with the expected impact and the reasoning behind it — never an unexplained directive.

Action catalog (illustrative)

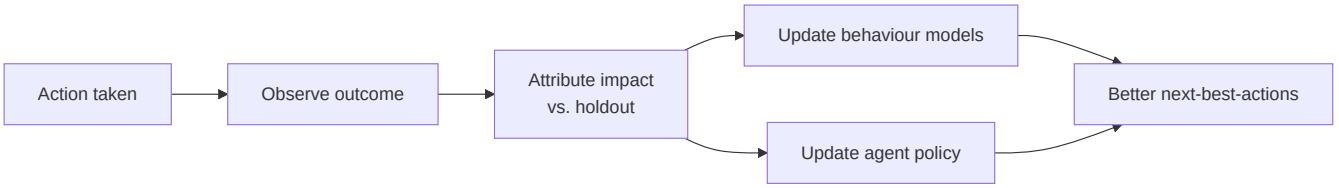
Action	Typical trigger
Targeted offer / promotion	High purchase propensity, price sensitivity known
Retention / save play	Rising churn risk on a valuable account
Onboarding nudge	Customer stalled at a known drop-off step
Cross-/up-sell	Affinity or usage signals readiness
Flag for human	High-stakes or low-confidence situation
Segment / audience update	Behavioural segment membership changed
Suppress / do nothing	Contact fatigue or no positive expected value

> **Doing nothing is a decision**
>
> Suppressing an action (to avoid fatigue or negative value) is a first-class output, not the absence of one.

Measuring outcomes (attribution)

A decision isn't finished when the action fires — it's finished when its effect is measured and fed back.

- **Experimentation** where feasible: holdouts / A-B tests so impact is causal, not just correlational.
- **Attribution**: tie outcomes (purchase, retention, engagement lift) back to the action that drove them.
- **Learning loop**: results update the behaviour models and the agent's sense of what works — so recommendations improve over time.



Human-in-the-loop and trust

Autonomy is **graduated** (see guardrails):

- Low-stakes, high-confidence, reversible actions → the agent may act automatically.
- High-stakes, low-confidence, or costly actions → the agent *recommends*, a human approves.
- Every decision — auto or approved — is logged with its reasoning for audit and review.

This is how the product earns the right to act more autonomously over time: by building a track record its operators can see and trust.

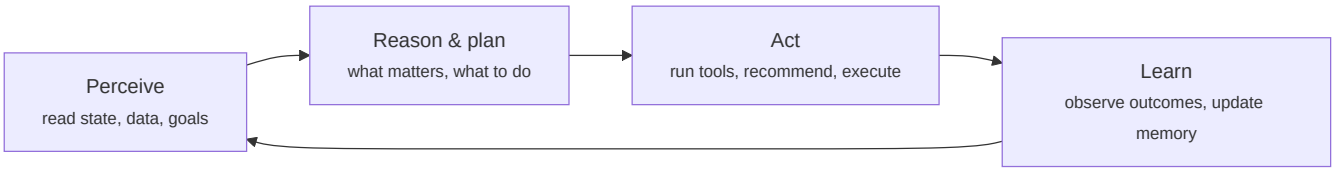
Open questions (for specs)

- How goals/objectives are expressed and weighted (single metric vs. multi-objective).
- The experimentation framework (how holdouts are managed and measured).
- Default autonomy levels per action type and how they're configured.
- How conflicting recommendations across segments/goals are reconciled.

The AutoAgent Loop

Pillar 5. The orchestrator that makes the system an *agent* rather than a pipeline. It holds goals, decides what to do next, invokes the other pillars as tools, remembers, and acts within guardrails.

The loop



Unlike a dashboard that waits for a query, the AutoAgent runs this loop on its own cadence — scheduled, event-triggered, or on demand — and chooses its own next question.

Perceive

Read current goals, recent data and events from the integration layer, outputs from the behaviour models, and its own memory of what it has already done and learned.

Reason & plan

Decide what matters now and form a plan: which segments or customers to look at, which insights to pursue, which decisions to prepare. This is where goals and priorities shape attention.

Act

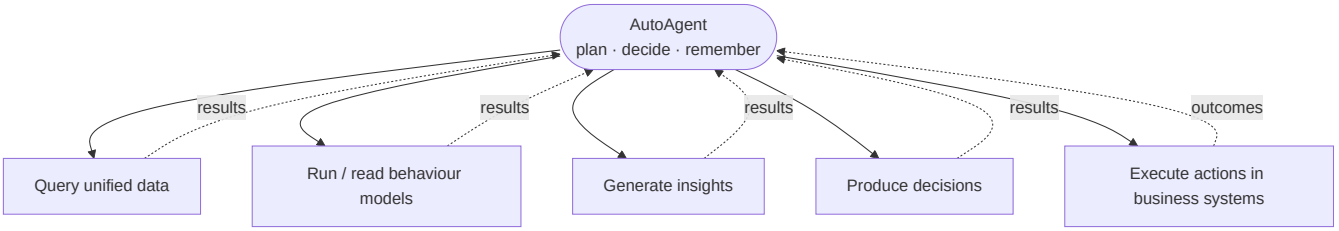
Invoke pillars 1–4 as **tools**, surface insights, propose or (within guardrails) execute actions. Every act is recorded with its reasoning.

Learn

Observe outcomes (via attribution), update memory and the sense of what works, and feed results back to the models. The next loop is better-informed than the last.

The agent as a tool-user

The AutoAgent doesn't reimplement the pillars — it *calls* them, plus the business's systems, as tools:



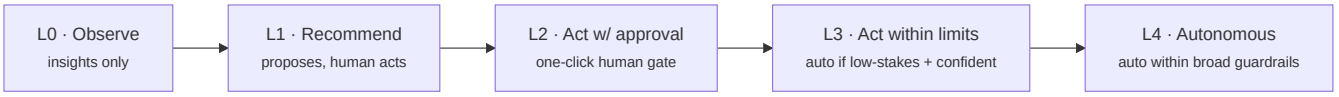
Memory

The agent is stateful. Memory makes it improve and keeps it from repeating itself:

Memory type	Holds	Used for
Working	The current investigation's context	Completing the task at hand
Episodic	Past insights, actions, and their outcomes	Not re-surfacing known findings; learning what worked
Semantic	Durable knowledge about the business, its customers, and its goals	Grounding reasoning in context

Guardrails and autonomy levels

Autonomy is **configurable and graduated**. Guardrails define what the agent may do on its own.



Typical guardrails:

- **Stakes & reversibility** — irreversible or high-cost actions require approval.
- **Confidence floors** — act only above a confidence threshold.
- **Budget & frequency caps** — spend and contact limits per customer/segment.
- **Excluded actions & audiences** — hard "never do this" rules.
- **Escalation** — anything outside the envelope is escalated to a human.

A business starts low (observe/recommend) and raises autonomy per action type as the agent earns trust through its track record.

Safety and accountability

- **Full audit trail** — every perception, decision, and action is logged with reasoning and provenance.
- **Explainability on demand** — any output can be traced back to its evidence.
- **Reversibility & kill-switch** — actions are reversible where possible, and the agent can be paused instantly.
- **Resistant to bad input** — the agent treats external/customer-supplied text as data, not instructions, and validates surprising signals before acting.

Open questions (for specs)

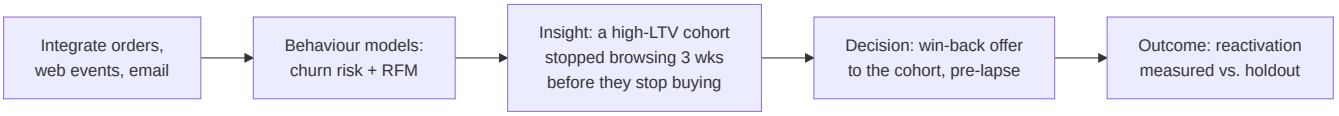
- The agent framework / reasoning model and how planning is implemented.
- Cadence: schedule, event triggers, and how continuous "continuous" is.
- Exact memory representation and retention.
- Default guardrail envelopes shipped out of the box.

Use Cases

Concrete scenarios that make the pillars tangible. Each follows the same arc: **integrate → model → insight → decision → outcome**.

1. E-commerce: rescue a silently churning cohort

Situation. An online retailer's repeat-purchase rate is slipping, but the headline numbers look fine.



- **Insight:** customers who were worth the most show a 3-week engagement drop *before* their purchases stop — a leading indicator no dashboard was tracking.
- **Decision:** the agent proposes a targeted, time-boxed win-back to exactly those accounts, before they lapse.
- **Outcome:** reactivation rate measured against a holdout; what works feeds the next round.

2. SaaS: turn usage signals into expansion

Situation. A B2B SaaS wants to grow existing accounts but can't tell which are ready.

- **Integrate:** product usage events + CRM + billing.
- **Model:** feature-adoption depth, engagement trend, expansion propensity.
- **Insight:** accounts that adopt feature *X* within 30 days expand at 3× the rate; a set of current accounts just crossed that threshold.
- **Decision:** flag those accounts to sales with the evidence and a suggested expansion play; nudge stalled accounts toward feature *X*.
- **Outcome:** expansion revenue attributed to flagged vs. unflagged accounts.

3. Subscription / DTC: reduce involuntary and voluntary churn

- **Integrate:** subscriptions, payments, support, app usage.
- **Model:** churn risk split by *cause* (price, dunning/failed payment, dissatisfaction, low usage).
- **Insight:** a third of churn is involuntary (failed payments), not a loyalty problem at all.
- **Decision:** route involuntary-risk customers to smart dunning; route dissatisfaction-risk to a save play; leave content customers alone.
- **Outcome:** churn reduction by cause, each measured separately.

4. Retail / omnichannel: unify online + in-store, then personalize

- **Integrate:** POS + e-commerce + loyalty, resolved to one customer.
- **Model:** cross-channel behaviour, product affinity, channel preference.
- **Insight:** customers who shop both channels are worth 2× single-channel ones, and a large single-channel group looks ready to cross over.
- **Decision:** encourage the cross-over with channel-appropriate, affinity-based offers.
- **Outcome:** cross-channel conversion and LTV lift.

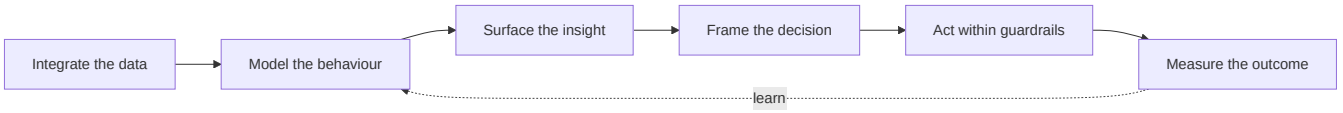
5. Plain-language investigation (pull mode)

Not every use case is automated. An operator can simply ask:

> *"Why did conversion drop last week?"*

The agent perceives the question, queries the integrated data, compares cohorts, and returns a **grounded insight** — "the drop is isolated to mobile users on channel Z after the checkout change on the 12th; desktop is flat" — with the evidence and a suggested next step. Same machinery, human-initiated.

Pattern across all use cases



The value isn't any single model or chart — it's an autonomous loop that keeps finding the non-obvious and tying it to a decision, across the whole customer base, continuously.

> **These are illustrative**
>
> Numbers and scenarios here are examples to convey the shape of the product, not claims or benchmarks. Real validation comes with the roadmap.

On-Prem, Local AI & Hardware

AutoAgentAI is built so it can run **entirely on the dealership's own premises**, on local hardware, with **local AI models**, so that **customer data never leaves the building**. This is not a bolt-on — it is a first-class deployment mode, and for most dealerships it is the *recommended* one, because the data involved is exactly the kind that privacy law treats as sensitive (see Privacy, PII & Future-Proofing).

> **The core promise**
>
> Everything the agent needs — the models that read and reason, the knowledge base, the customer data, the decision-making — can live on a machine in the back office. The only things that cross the network are the things you explicitly choose to send out (e.g. a manufacturer feedback summary), and even those are on your terms.

Why on-prem is now realistic

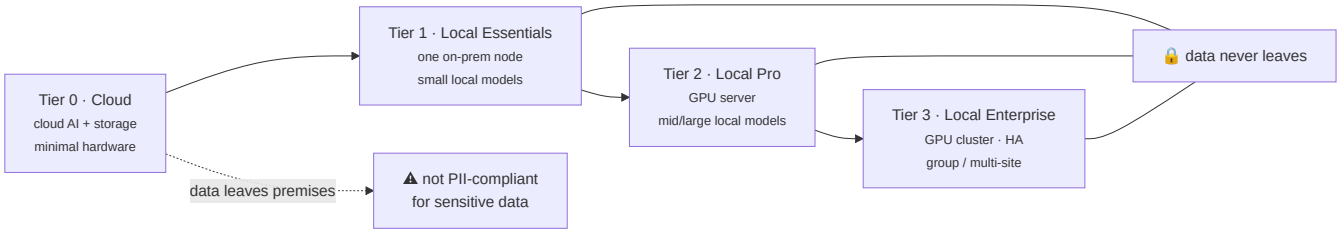
Two things changed:

1. **Open-source local models got good enough.** The open models you can run on a single workstation today do the specific jobs this product needs — reading notes, classifying intent, summarizing, extracting structure, ranking options, drafting language — at a quality that was cloud-only a short time ago. See Local Models Are Enough.
2. **Agentic AI matured.** The orchestration, tool-use, and memory patterns that make an autonomous agent work no longer require a frontier cloud model for every step. The heavy reasoning can be reserved, batched, or done locally.

Together, these make a fully local, private, autonomous customer-intelligence agent *buildable on dealership hardware* — which a few years ago it was not.

The four deployment options

AutoAgentAI ships as **four tiers**, from cheapest-but-not-private to fully sovereign. A dealer picks based on how sensitive their data is, what they can spend, and how much they want to keep in-house.



	Tier 0 — Cloud	Tier 1 — Local Essentials	Tier 2 — Local Pro	Tier 3 — Local Enterprise
AI	Cloud AI services	Small/mid local models	Mid/large local models	Largest local models, redundant
Storage	Cloud	On-prem	On-prem	On-prem, replicated
Hardware	None / minimal	One mini-server or workstation	GPU server/workstation	GPU cluster, HA, UPS
Data location	Third-party cloud	**On premises**	**On premises**	**On premises (sovereign)**
PII-law posture	⚠ Not compliant for sensitive data	Compliant	Compliant	Compliant + auditable at scale
Cost	Lowest upfront	Low	Moderate (capex)	Highest (capex)
Best for	Pilots, low-sensitivity, proof-of-value	Single rooftop, getting started privately	A full dealership operation	Dealer groups, flagship, multi-site
Scale	Elastic	1 site, light load	1 site, full load, many nodes	Many sites, high concurrency

Tier 0 — Cloud (cheap, fast, **not** PII-compliant)

Uses cloud AI services and cloud storage. **No or minimal on-site hardware.** It is the cheapest and quickest way to stand the system up and prove it works — and it **can genuinely work**. The catch is unavoidable: customer data leaves the premises and passes through third-party cloud and AI providers, so it is **not suitable for PII-regulated customer data**. Use it for pilots, demos, non-sensitive workloads, or proving value before investing in hardware — with eyes open about the trade-off (see Cloud vs On-Prem: The Risks).

Tier 1 — Local Essentials (entry on-prem, private)

A single on-prem node — a capable mini-server or workstation — runs small-to-mid **local** models and holds the data locally. **Data never leaves the building.** Modest cost, modest footprint; ideal for a single rooftop that wants privacy from day one and a real but contained workload.

Tier 2 — Local Pro (dealership-grade, private)

A proper on-prem **GPU server/workstation** runs mid-to-large local models with a local knowledge base and vector store, driving a full fleet of kiosks and iPads across the dealership. Handles the real concurrent load of a busy showroom. This is the sweet spot for most single dealerships that take their data seriously.

Tier 3 — Local Enterprise (group / flagship, sovereign)

A **GPU cluster** with high availability, redundancy, and the largest local models, serving a dealer group or multiple sites. Data sovereignty across the whole group, auditable at scale, with edge-sync between locations. For organizations where the data is strategic and downtime is unacceptable.

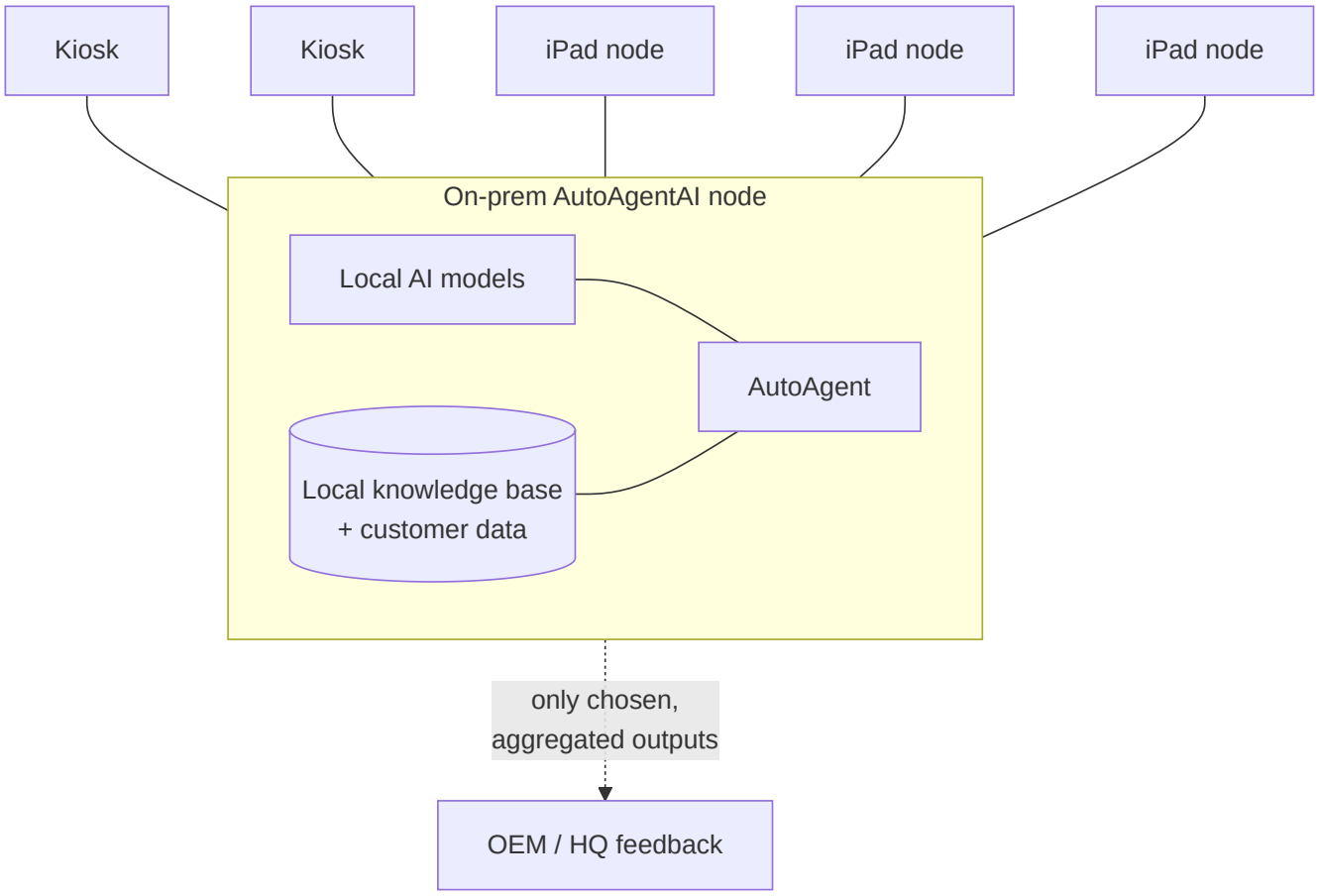
> **Same product, different engine**
>
> All four tiers run the *same* AutoAgentAI — same pillars, same interfaces, same agent. What changes is **where the compute and data live** and **how much** of it you can run at once. A dealer can start at Tier 0 to prove value, then move the exact same workflows on-prem at Tier 1–3 when ready.

Interfaces: kiosks and iPads

The system meets people where they are, with two complementary interfaces:

- **McDonald's-style self-serve kiosks.** Large, friendly, touch-first terminals — the same pattern customers already trust from quick-service restaurants. Low intimidation, self-paced, and they capture rich interaction data as a natural by-product of being genuinely useful to the customer.

- **iPads.** The mobile, handheld form of the same system. Crucially, **iPads are generally the *nodes* that connect to the whole system** — a salesperson or greeter carries one, and through it the whole local AutoAgentAI (models, data, agent) is at hand anywhere on the lot or in the showroom.



The iPads and kiosks are **thin** — the intelligence and the data stay on the on-prem node. A lost or stolen iPad carries no customer database; it is a window into the local system, not a copy of it.

What a local deployment contains

Component	Role	Lives where
Local AI models	Reading, reasoning, extraction, drafting, ranking	On-prem node (GPU)
Knowledge base / vector store	The dealership's customers, vehicles, history	On-prem storage
Unified customer profiles	The resolved view per customer	On-prem storage
AutoAgent runtime	Orchestration, memory, guardrails	On-prem node
Kiosk + iPad apps	Thin interfaces / nodes	Devices on the LAN
Outbound (optional, chosen)	Aggregated OEM feedback, group sync	Egress on your terms

Hardware guidance (illustrative)

Exact specifications are a Phase-1 spec decision and depend on chosen models and load. As rough orientation only:

- **Tier 1:** a single modern workstation with one capable GPU and ample RAM/SSD — enough for small/mid local models and a single site's light, interactive load.
- **Tier 2:** a dedicated GPU server (one or more workstation/data-center GPUs), fast NVMe storage, and a UPS — enough for larger models and a showroom's full concurrent kiosk + iPad load.
- **Tier 3:** multiple GPU servers in a rack with redundancy, high-availability, and replicated storage — for dealer-group scale and uptime guarantees.

> **Numbers here are orientation, not a quote**
>
> Treat all hardware and cost statements as directional. Final sizing comes with the specifications once models and expected load are fixed.

Next: Privacy, PII & Future-Proofing · Cloud vs On-Prem: The Risks · Local Models Are Enough

Privacy, PII & Future-Proofing

AutoAgentAI works with the most sensitive data a dealership holds: who its customers are, what they own, what they can afford, what they want, and how they behave. That is **personally identifiable information (PII)**, and in many cases special-category or financial data on top. The product's privacy stance is therefore not a feature — it is a precondition.

> **Not legal advice**
>
> This page explains the design posture and the principles it is built around. It is not legal advice. Each dealer's obligations depend on their jurisdiction and should be confirmed with counsel. The point of the on-prem design is to make compliance *easier to achieve and prove*, whatever the specific rules.

The regulatory landscape (why this matters)

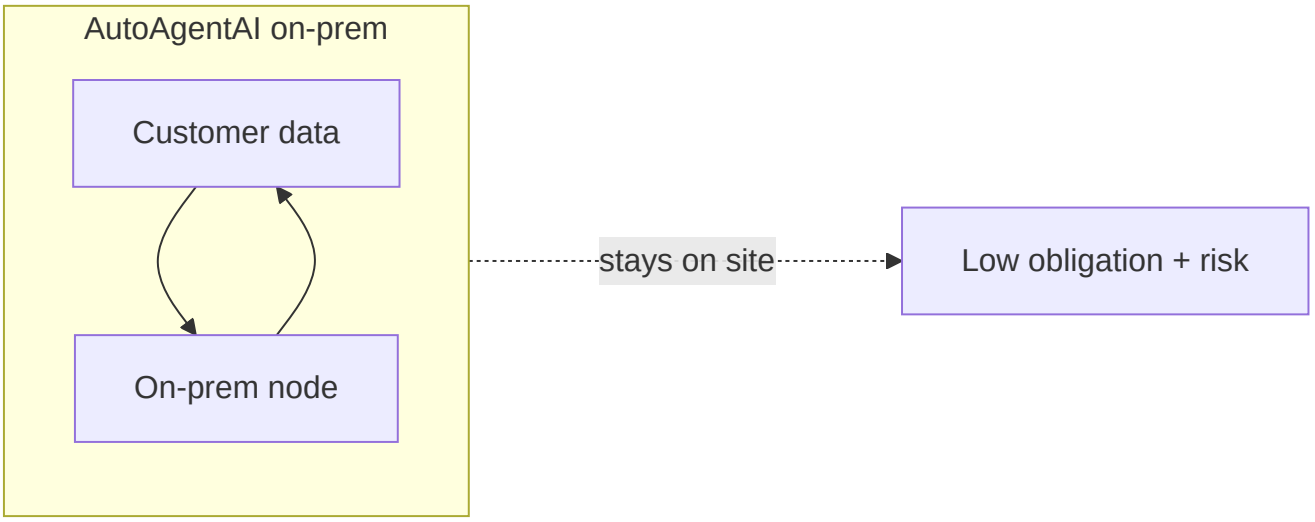
Customer data is governed by a growing, tightening web of privacy law. A non-exhaustive map of what dealerships touch:

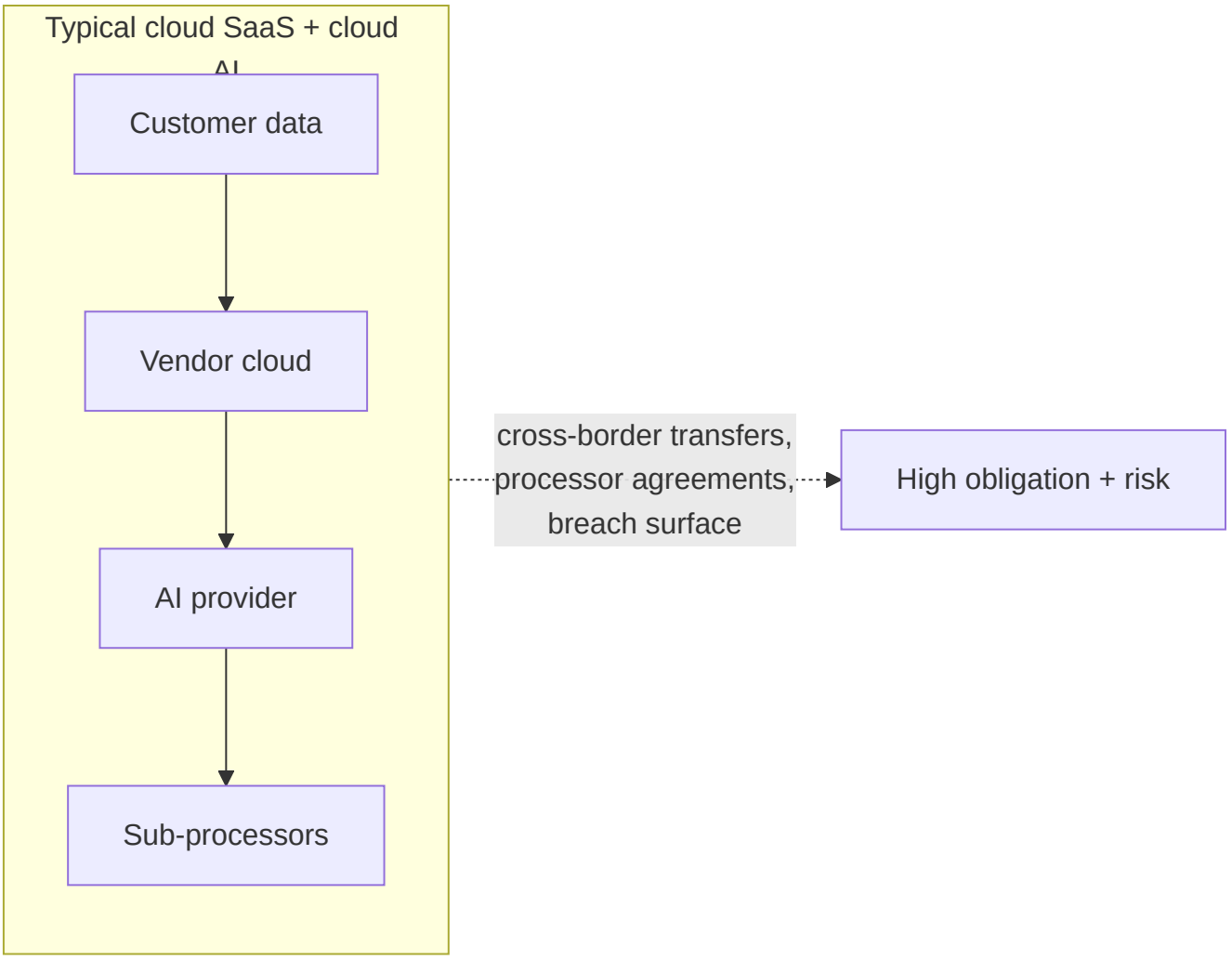
Regime / area	Who/what it covers	Core demands (simplified)
GDPR (EU/EEA)	Any EU resident's personal data	Lawful basis, consent, minimization, access/erasure rights, breach reporting, limits on transfers
CCPA / CPRA (California)	California consumers	Disclosure, opt-out of sale/sharing, deletion rights, sensitive-PI limits
PIPEDA (Canada) / provincial laws	Canadian individuals	Consent, purpose limitation, access, safeguards
State privacy laws (US)	A growing list of states	Access, deletion, opt-out, data-minimization echoes of CPRA
Financial data rules	Credit apps, financing	Heightened safeguards, Gramm-Leach-Bliley-style handling in the US
Biometric / image laws	Camera/vision data (e.g. BIPA)	Often require explicit consent and strict handling

The direction of travel is one-way: **more** rights for individuals, **more** restrictions on moving and selling data, **more** scrutiny of AI and automated decisions. A system designed around last year's rules ages badly.

The design posture: data that doesn't leave can't leak

The single most powerful privacy move is **not moving the data**. AutoAgentAI's on-prem tiers (see On-Prem AI & Hardware) keep customer data and the AI that reads it **inside the dealership**. This collapses whole categories of risk and obligation:





- **No third-party processors to paper over.** No chain of cloud vendors and AI-provider sub-processors each needing contracts, assurances, and audits.
- **No cross-border transfer problem.** Data that never leaves the premises never crosses a border.
- **A tiny breach surface.** You cannot leak from a cloud you are not using.
- **Honest answers to customers.** “Your information stays here, on our systems, and is used to serve you” is both a compliance posture and a trust-builder.

Principles built in

The product is designed around the principles the laws keep repeating, so that compliance is the default rather than a scramble:

1. **Data minimization.** Collect and retain only what a decision needs.
2. **Purpose limitation & consent.** Each customer record carries what they consented to; the agent only uses and acts within that consent.
3. **Individual rights are operational.** Access, correction, and **erasure** requests are first-class actions the system can execute, with proof.
4. **Provenance & auditability.** Every insight and action traces to its data and reasoning — so you can show *why* a decision was made, which regulators increasingly require for automated decisions.
5. **Security by default.** Encryption, least-privilege access, and access logging — far simpler to guarantee on a box you control than across a vendor chain.
6. **Human accountability.** High-stakes or automated decisions stay explainable and reversible, with a human in the loop where it matters.

Future-proofing

“Future-proof” here means specific, testable things:

- **Sovereignty by default.** Because data stays on-prem, most *new* transfer, localization, and sale-of-data rules land as no-ops — you already aren’t doing the risky thing.
- **Model independence.** Local, open models mean you are not hostage to a cloud provider changing terms, prices, retention policies, or training practices. If a better open model appears, you swap it in; your data and workflows don’t move.
- **Auditable AI.** As rules on automated decision-making tighten, the provenance trail is already there to demonstrate fairness, basis, and human oversight.
- **Portability.** The same AutoAgentAI runs across all four tiers, so a dealer can tighten posture (Tier 0 → Tier 3) as obligations grow, without re-platforming.
- **Consent that travels forward.** Consent and preferences are modeled explicitly, so new consent requirements extend the model rather than breaking it.

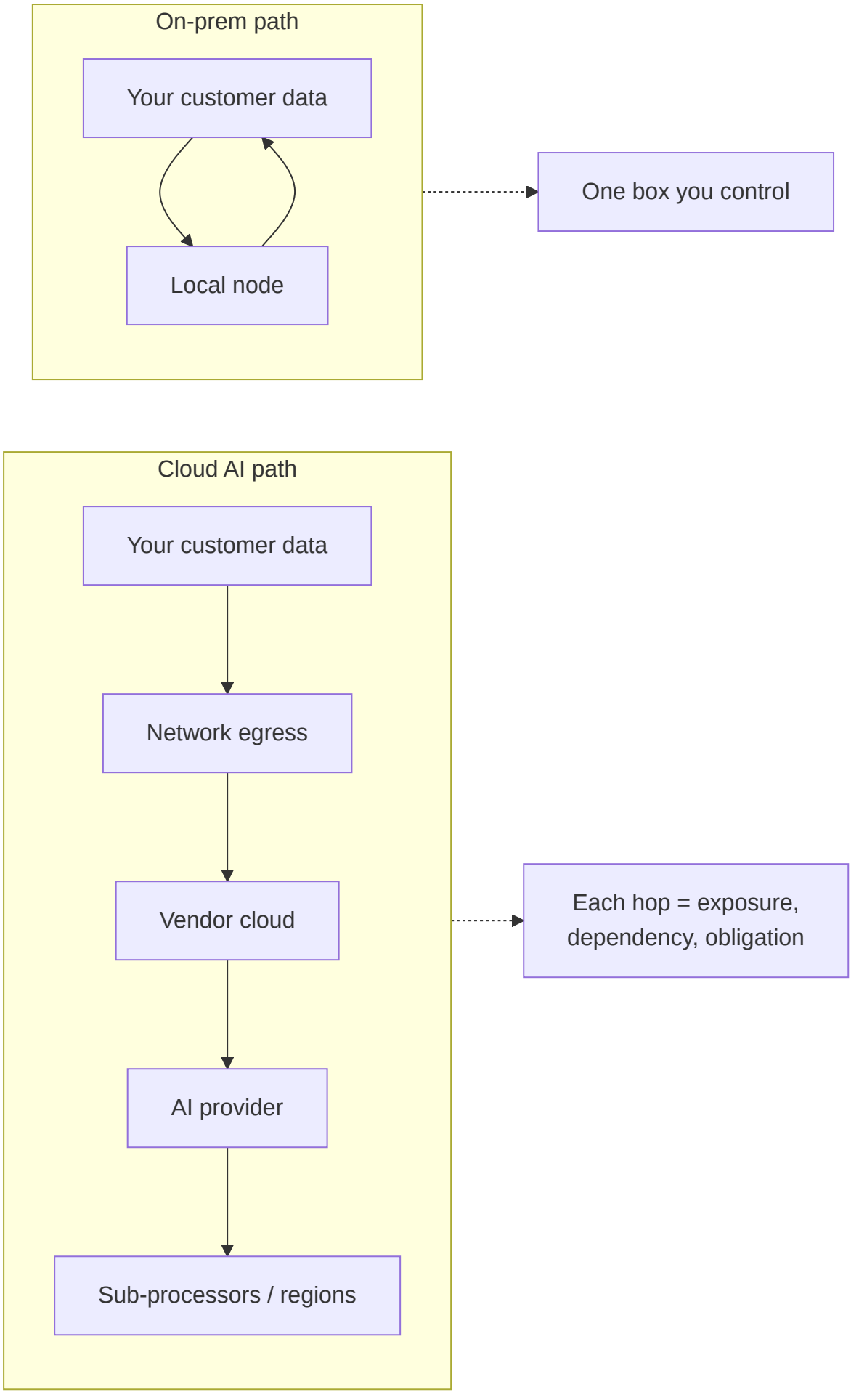
> **The strategic read**
>
> Privacy law is a moving target that only moves in one direction. The cheapest way to keep up with a tightening rule-set is to **not depend on the behaviours those rules restrict** — moving, selling, and exposing customer data. On-prem, local-model AutoAgentAI is, in effect, a bet that sovereignty is the durable position. See Cloud vs On-Prem: The Risks for the other side of that bet.

Cloud vs On-Prem: The Risks

Tier 0 (cloud AI services + cloud storage) is cheap, fast, and genuinely works — and for sensitive customer data it carries risks that on-prem simply removes. This page lays the two side by side so the trade-off is made with eyes open, not by default.

> **The one-sentence version**
>
> With cloud AI you are renting someone else's computer to hold and read your most valuable, most regulated data; with on-prem you own the box, the model, and the risk surface — and the data never leaves.

Where the risk actually lives


On-prem path
Your customer data
Local node
One box you control
Cloud AI path
Your customer data
Network egress
Vendor cloud
AI provider
Sub-processors / regions
Each hop = exposure, dependency, obligation

Every arrow leaving your building is a place where data can be exposed, retained, subpoenaed, repriced, or quietly used for something you didn't intend.

The risks of cloud AI for this data

Risk	What it means for a dealership
Data leaves your control	Customer PII, finances, and behaviour sit on infrastructure you don't own and can't fully inspect.
Secondary use / training	Depending on terms, prompts and data may be logged, retained, or used to improve vendor models. Terms can change.
Vendor lock-in & pricing power	Your workflows depend on an API someone else prices, throttles, deprecates, or changes.
Availability dependence	An outage or a suspended account at the provider takes your showroom tools down with it.
Breach surface	A breach anywhere in the vendor → AI-provider → sub-processor chain can expose your customers, and it's your name on the apology.
Compliance drag	Processor agreements, transfer mechanisms, audits, and disclosures — ongoing paperwork and liability for every vendor in the chain.
Cross-border exposure	Data may be processed in regions whose laws (and access powers) differ from yours.
Loss of provenance	Harder to prove *why* an automated decision was made when the reasoning happened in a black-box API.
Confidential leakage	Your competitive edge — how you sell, what works — flows through a third party.

The risks on-prem removes (and the ones it adds)

Removed or sharply reduced: data leaving the premises, secondary use by a vendor, cross-border transfer, dependence on a provider's pricing/uptime/terms, and most of the processor-chain compliance drag.

Added (and manageable):

On-prem cost	Mitigation
Upfront hardware (capex)	Scoped by tier; amortizes against avoided per-call cloud fees and risk
You run the box	Managed appliance model, updates, and support; thin kiosks/iPads keep it simple
You own uptime	UPS, and HA/redundancy at Tier 3
Model updates are yours to apply	Open models are swappable; updates are a controlled, local step

On-prem trades a *recurring, external, hard-to-bound* risk (someone else holds your data) for a *one-time, internal, boundable* cost (you run a server). For data this sensitive, that trade usually favors on-prem.

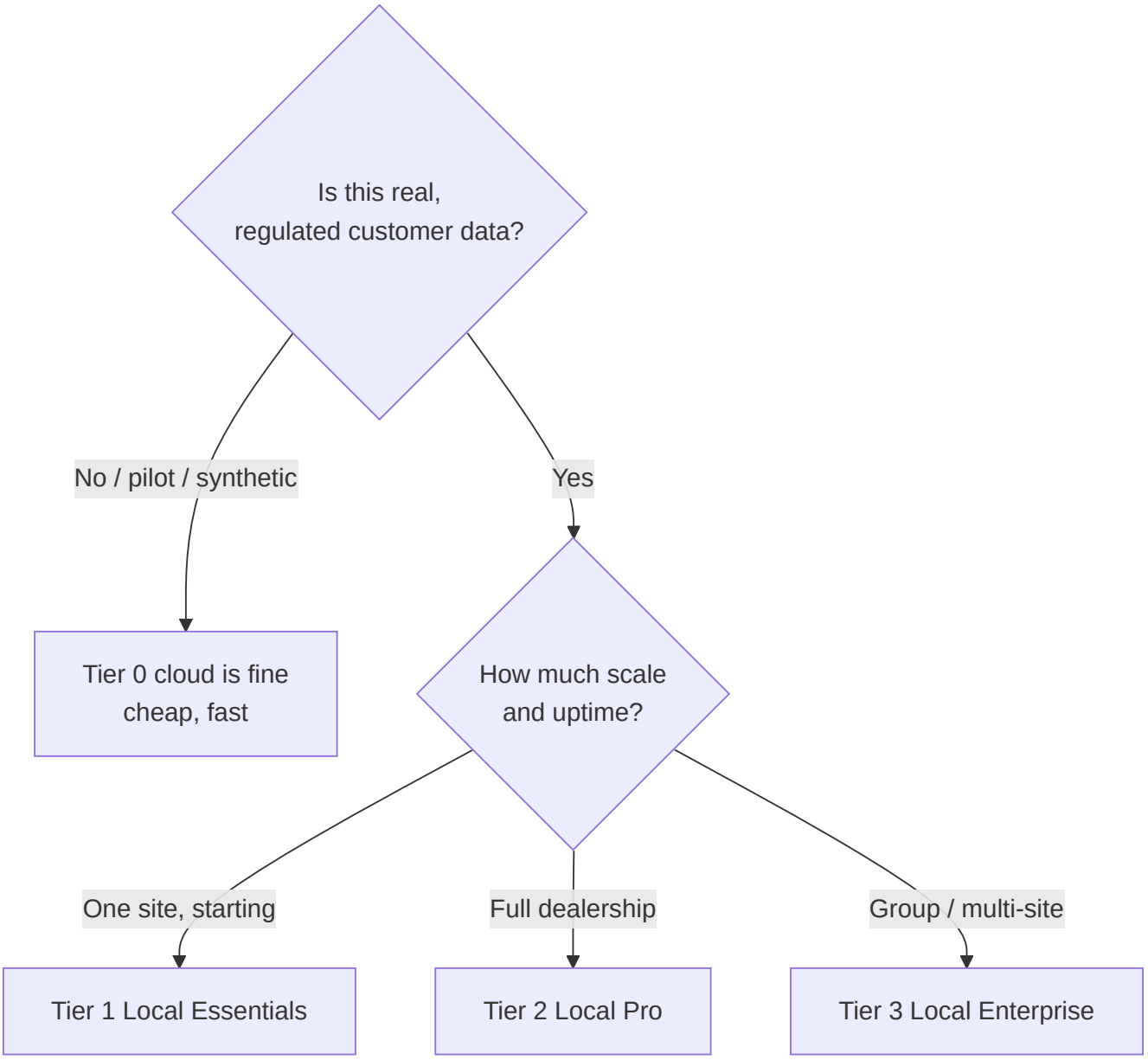
When cloud (Tier 0) is still the right call

Tier 0 is not wrong — it's wrong *for sensitive data*. It's a good fit when:

- You're running a **pilot** or demo to prove value before buying hardware.
- The workload uses **non-sensitive or synthetic** data.
- You explicitly accept the posture for a bounded, short-lived purpose.

The product is designed so you can **start on Tier 0 and migrate the identical workflows on-prem** once value is proven — no rebuild. See On-Prem AI & Hardware.

The decision, framed



The asymmetry that matters

A cloud cost saving is visible and small each month. A privacy failure with customer PII is invisible until it isn't, and then it is large, public, and permanent. The risks are not symmetric — which is why, for this data, the default recommendation is on-prem. See Privacy, PII & Future-Proofing.

Local Models Are Enough

Thesis: for the specific jobs AutoAgentAI needs an AI model to do, today's open-source **local** models do the work as well as the big cloud models — well enough that giving up the cloud costs you nothing that matters, and buys you everything on the privacy and risk pages.

> **Why this is true now and wasn't before**
>
> The gap between frontier cloud models and the best open models has narrowed to the point where, on *bounded, well-scoped tasks*, they are interchangeable. Our tasks are exactly that kind — bounded and well-scoped — not open-ended frontier reasoning. You don't need the world's biggest model to read a sales note.

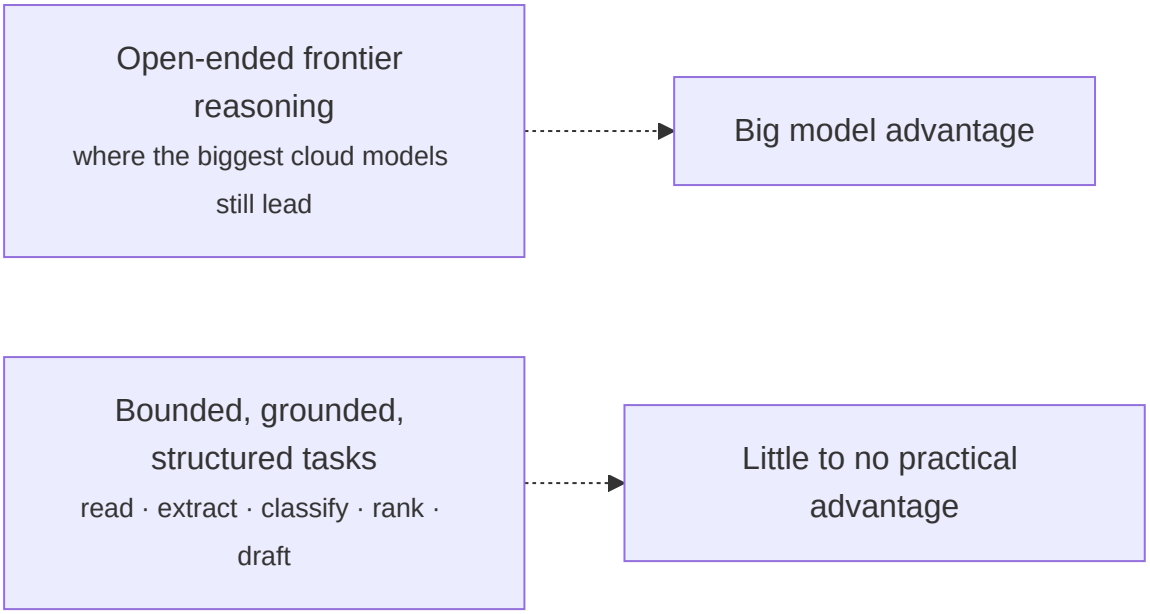
What the model actually has to do here

The agent doesn't need a model that can do everything. It needs a model that can do *these* things, reliably, on dealership data:

Job	What it looks like	Difficulty for a local model
Read & understand notes, messages, reviews	"Customer worried about resale value, has two kids"	Easy — well within local models
Extract structure from free text	Pull intent, budget signals, objections into fields	Easy / moderate
Classify & tag	Intent, sentiment, lifecycle stage, segment	Easy
Summarize	Condense a visit or a history into a brief	Easy
Rank & choose	Next-best-action from a known menu	Moderate — and aided by the models in Behaviour Models
Draft language	A friendly follow-up, a kiosk prompt	Easy
Reason over retrieved facts	Answer grounded in the local knowledge base	Moderate — handled well with retrieval

None of these is a frontier task. They are the bread-and-butter capabilities that even modest open models now handle, especially when **grounded in retrieval** (the model reasons over *your* facts rather than from memory) and **constrained to structured outputs** (it fills fields, not freeform essays).

Why scoped tasks favor local models



Three design choices shrink the model's burden until a local one is plenty:

1. **Retrieval-grounded.** The knowledge lives in the local knowledge base; the model reads from it instead of recalling. This leans on comprehension, which local models are good at — not on a giant parametric memory.
2. **Structured outputs.** The agent asks the model to populate fields and pick from menus, not to free-associate. Constrained tasks are where small models are most reliable.
3. **The agent does the planning.** Multi-step logic lives in the agent loop and the deterministic tools — not in one giant model call. Each model call is small and well-defined.

Honest limits and how we handle them

Being truthful about where local models are weaker, and the mitigations:

Limitation	Mitigation
Hardest open-ended reasoning lags frontier models	We don't route such tasks to the model; the agent decomposes them
Smaller models hallucinate more when ungrounded	Retrieval-grounding + "answer only from provided facts" + provenance checks
Throughput is bounded by your GPU	Tier sizing (hardware); batch non-interactive work
Occasional task needs more horsepower	Optional: a larger local model at Tier 2/3, or an explicit, consented, non-PII cloud call for that one task

The economics flip, too

Cloud models charge per call, forever, and the bill grows with success. A local model is a **fixed cost** you've already paid: once the hardware is in, inference is effectively free and unlimited, which changes how the agent can behave — it can afford to be *continuously* curious (the whole point of an autonomous agent) without a meter running on every thought.

	Cloud models	Local models
Cost shape	Per-call, recurring, scales with use	Fixed capex, then ~free inference
"Always-on curiosity"	Expensive — every loop costs money	Cheap — run the loop as often as you like
Data exposure	Leaves premises	Stays on premises
Dependency	Vendor terms/pricing/uptime	Yours

The conclusion

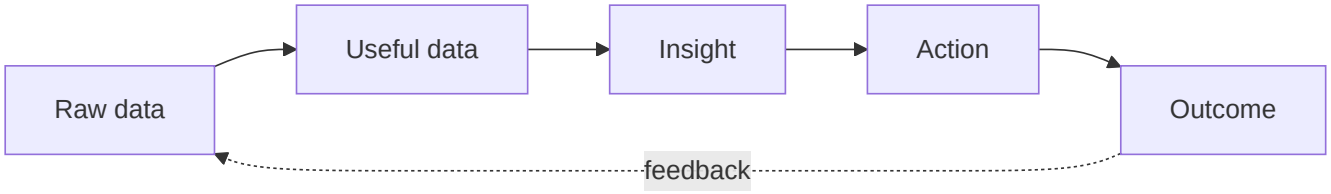
> For AutoAgentAI's actual workload — reading, extracting, classifying, summarizing, ranking, and drafting over a dealership's own grounded data — **a local open-source model does the job as well as a cloud model**, at a fixed cost, with the data never leaving the building. The cloud's remaining edge is in open-ended frontier reasoning the agent is designed not to depend on.

That is what makes the on-prem tiers not a compromise but the *better* engineering choice for this problem.

> **Claims here are directional**
>
> Specific model choices and measured quality bars are a Phase-1 specification task — including the evaluation suite that proves, on real dealership tasks, that the chosen local model clears the bar.

From Raw Data to a Decision: A Dealership Scenario

Every claim in these docs comes down to one motion: **raw data → useful data → insight → action → outcome.** This article walks that motion end to end, with a single concrete dealership scenario, so the abstract pillars become something you can picture happening on a Tuesday afternoon.



The four transformations

Stage	What it is	What it takes
Raw data	Everything that happened, unstructured and scattered	Capture (kiosks, iPads, systems)
Useful data	The same events, unified and resolved per customer	Customer Integration
Insight	A grounded, tailored finding that implies a decision	Insights Engine + Behaviour Models
Action	A specific, measured next-best-action	Decision Intelligence

The scenario: meet “Maria”

1 · Raw data (what actually happened)

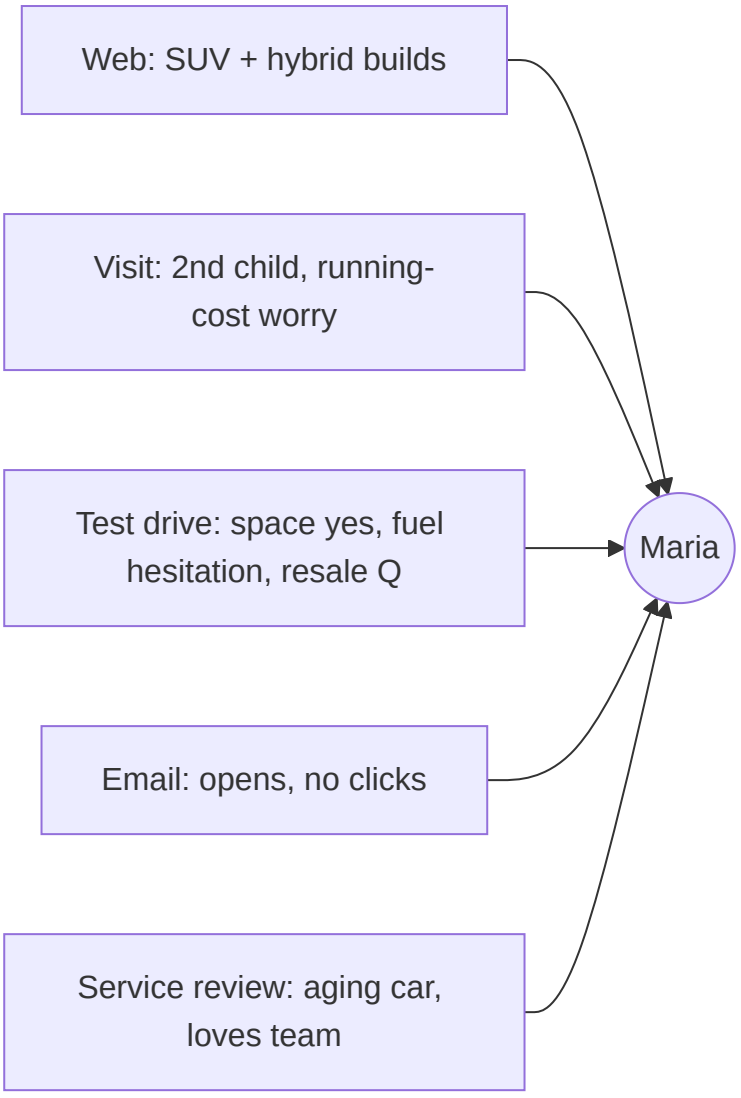
Over six weeks, Maria generates signal nobody is joining up:

- Browses the dealership site three evenings, lingering on a mid-size SUV and a hybrid trim; configures a build but doesn’t submit.
- Walks in on a Saturday. A greeter with an **iPad** checks her in; she mentions she has “a second kid on the way” and is “worried about running costs.”
- Test-drives the SUV. The salesperson’s notes (dictated into the iPad): *“liked space, hesitated on fuel, asked about resale.”*
- Opens two follow-up emails but clicks neither offer.
- Leaves a 4-star service review on her *current, aging* car two weeks later: *“love the team, car’s getting tired.”*

Today, those five things live in five systems and tell no one a story.

2 · Useful data (unified and resolved)

AutoAgentAI’s integration layer resolves all five to **one customer, Maria**, on-prem:



Now it's not five fragments — it's a person with a trajectory.

3 · Insight (grounded and tailored)

The behaviour models and insights engine read the unified picture and surface a *custom insight*, not a generic metric:

> **Insight**
>
> **Maria is a high-propensity hybrid-SUV buyer whose single blocker is running cost, and whose current car is near end-of-life.** Customers matching this pattern (growing family + fuel hesitation + aging trade-in + service loyalty) convert at well-above-average rates **when shown a total-cost-of-ownership comparison and a trade-in value up front** — and churn to a competitor when left to a generic follow-up. **Evidence:** her web/visit/drive/review signals + the dealership's own history of similar customers. **Confidence:** high. **Expected impact:** one incremental hybrid sale + a trade-in.

No salesperson assembled this. It was impossible for one to (see Why Salespeople Can't Capture This Data).

4 · Action (a specific next-best-action)

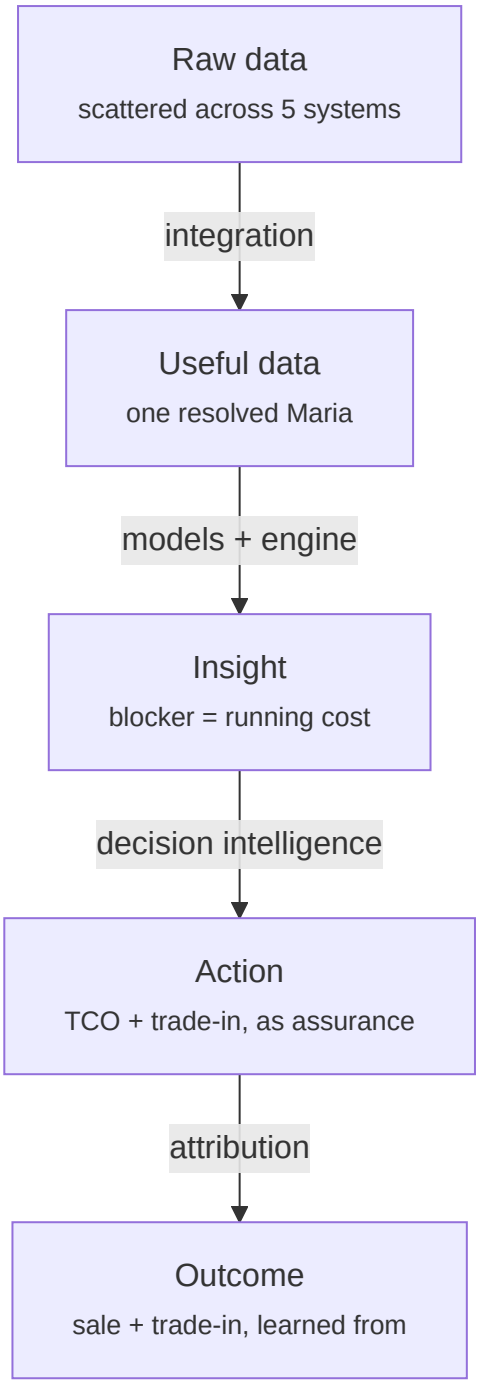
Decision Intelligence turns the insight into a concrete, guardrailed action — not a vague "follow up":

- **Next-best-action:** a personalized follow-up that leads with a **total cost-of-ownership** comparison (addressing the exact blocker) *and* an up-front **trade-in estimate** for her aging car.
- **Routed to** her salesperson's iPad with the reasoning and talking points, framed as *assurance*, not pressure (see Reinventing the Salesperson's Role).
- **Guardrails:** within contact-frequency limits and her consent; a human sends it.

5 · Outcome (measured, and fed back)

The action is tracked against a holdout. Maria books a second drive, trades in, and buys the hybrid. The **outcome feeds back**: the models learn that this pattern + this action worked, sharpening the next recommendation — and, in aggregate, this signal can flow to the OEM (see Closing the Loop to the OEM).

Why each stage needs the system



Remove any stage and the value collapses: raw data with no integration is noise; integrated data with no insight is a nicer database; an insight with no action is a chart nobody acts on; an action with no measured outcome never improves. The agent's job is to run this whole motion, continuously, for **every** customer at once.

> **Illustrative**
>
> "Maria" and the numbers are an illustration of the mechanism, not a case study or a performance claim.

Reinventing the Salesperson's Role

AutoAgentAI changes what a car salesperson is *for*. The parts of the job that make customers brace — the pressure, the information asymmetry, the chase — are exactly the parts the agent absorbs. What's left is the part customers actually want: **a knowledgeable human offering assurance and assistance.**

> **The shift in one line**
>
> From *persuading people who feel outgunned* to *helping people who feel understood.* The agent handles capture, memory, and analysis; the human handles trust, reassurance, and care.

What the job is today — and why it's adversarial

The traditional model puts the salesperson and the customer on opposite sides:

- The salesperson holds information the customer doesn't (cost, margin, what's flexible), creating an **asymmetry customers can feel**.
- Their incentive is to close *now*, which reads as **pressure**.
- They must *remember and chase* — juggling dozens of prospects from memory and sticky notes, inevitably letting people fall through the cracks.
- Much of their day is **data-entry and follow-up admin**, not helping anyone.

Customers show up defensive because the structure *makes* it adversarial. That defensiveness is bad for everyone, including the salesperson's close rate.

What the agent takes off their plate



The agent never forgets a customer, never loses a note, and quietly tells the salesperson *who needs what, and why, right now* — on their iPad. The salesperson walks up already knowing Maria is worried about running costs, so they can **lead with the answer to her actual concern** instead of a pitch.

From closer to concierge

Dimension	Traditional salesperson	AutoAgentAI-enabled host
Stance	Persuade / close	Assure / assist
Information	Asymmetry (they know more)	Shared (customer feels informed)
Memory	Human, lossy	Agent-backed, complete
Time spent	Admin, chasing, guessing	Present with the customer
What the customer feels	Pressured, defensive	Understood, helped
Incentive felt	"Sell me something"	"Help me decide well"

The role becomes closer to a **concierge or trusted advisor**: the person who makes a big, anxious purchase feel calm and well-handled. That is a *better* job — more human, less burnout, and, not incidentally, it converts better, because people buy from people they trust.

"Replace some salespeople" — honestly

It's fair to be direct: when one agent captures, remembers, and analyzes what used to take a floor of people chasing and guessing, a dealership **needs fewer people doing that low-value work.** But the framing matters:

- The work that disappears is the **admin, the chasing, and the guessing** — the parts nobody enjoyed and nobody was good at at scale.

- The work that remains and grows is the **human assurance** — which the agent can't do and customers value most.
- The net effect is a **smaller team of better-supported people doing higher-value work**, each one amplified by the agent, each customer getting more attention, not less.

> **The reframe**
>
> The agent doesn't replace the salesperson with a machine. It replaces the *adversarial, admin-heavy* salesperson with a *supported, trusted host* — and lets fewer of them serve more customers, better.

Why this is good for the customer, too

Buying a car is one of the most stressful consumer purchases there is. An experience where:

- the person helping you already understands your situation,
- isn't pressuring you because the "closing" machinery runs quietly in the background,
- and can answer your real worry with real numbers,

...is simply a better experience. The agent's job is to make the human in front of the customer **more human**, not to remove them. See the mechanism in From Raw Data to a Decision.

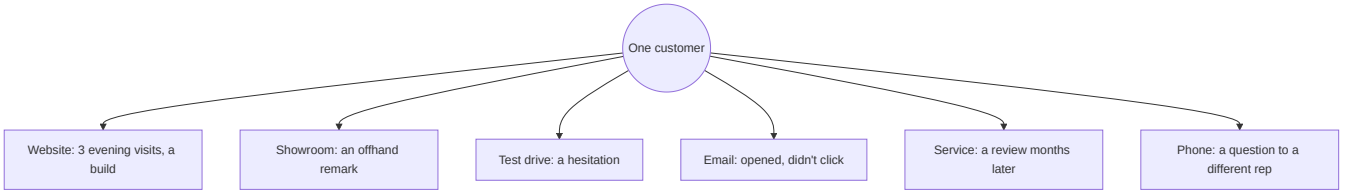
Why Salespeople Can’t Capture This Data

The data AutoAgentAI works with isn’t hard for a salesperson to capture by accident — it’s **impossible** for them to capture by design. Not because they’re not good at their jobs, but because the task is structurally beyond what any human can do. This matters, because it’s the reason this data has stayed uncaptured and therefore *unexploited* across the whole industry.

> **The core point**
>
> You cannot ask a human to be a perfect, tireless, unbiased sensor and analyst across hundreds of customers and thousands of tiny signals. The data isn’t missing because people are lazy — it’s missing because capturing it is not a human-scale task.

The signals are tiny, constant, and multi-channel

A single customer throws off signal across channels no salesperson can see at once:



The salesperson sees, at most, the showroom slice — and only while standing there. The evening web visits, the email behaviour, the service review, the call to a different rep: invisible to them. The story only exists when **all** of it is joined, and no human is in all those places.

Six reasons it’s impossible for a human

#	Barrier	Why it defeats a person
1	**Volume**	Hundreds of active prospects, each generating signal daily. No memory holds that.
2	**Simultaneity**	Signals arrive across web, phone, email, showroom, service — at the same time, in different places.
3	**Subtlety**	“Worried about resale,” a 20-second extra pause on a trim — the signal is in the small stuff.
4	**Recall decay**	Even captured, human memory of a conversation is gone or distorted within days.
5	**The capture tax**	Logging a detail means *not* being present with the customer. Doing the job well means *not* recording it.
6	**No cross-customer view**	A human can’t compare this customer to the 400 similar ones to know what usually works.

Barrier 5 is the cruel one: **the act of capturing data competes with the act of selling.** Every note typed is attention taken off the human in front of you. So the best salespeople, rightly, *don’t* capture — they stay present — and the data evaporates.

What actually happens today

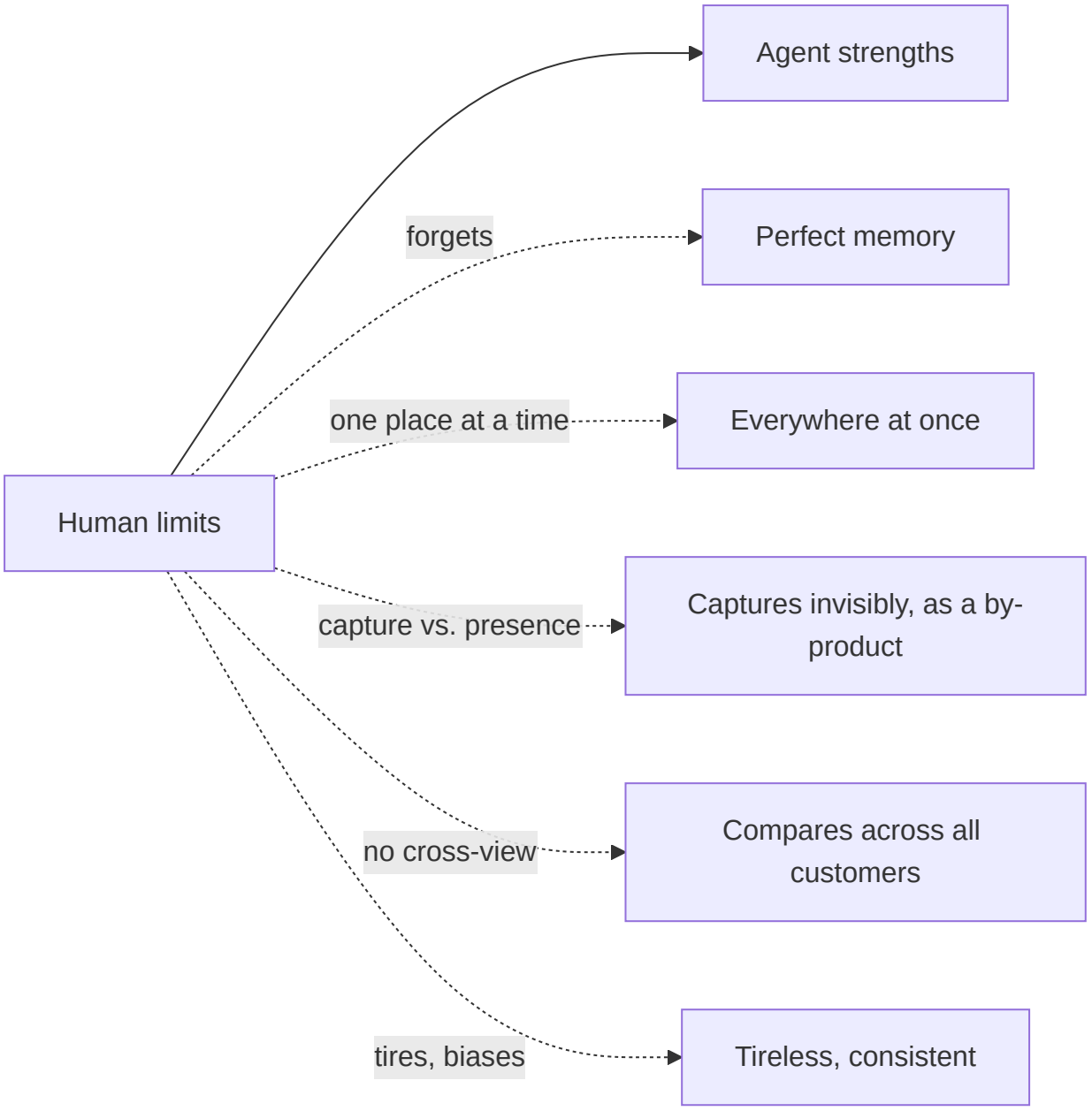
- A fraction of interactions get a terse CRM note, written later, from memory.
- The note captures what the rep *thought mattered*, not the full signal.
- Web, email, service, and call data sit in other systems, unjoined.

- When the rep leaves, their “book” of customer knowledge walks out the door.
- Cross-customer patterns — the most valuable thing — are never computed at all.

The result: the industry sits on an ocean of the most valuable behavioural data there is (see The Most Valuable Data in Autos) and captures a thimble of it, badly.

Why only an agent can

An autonomous agent is the only thing shaped right for this task, because it is everything a human can’t be here:



- It is **in every channel at once** — web, kiosk, iPad, email, service — because it *is* the system those run on.
- It **captures as a by-product** of being useful (the kiosks and iPads log rich signal just by serving the customer), so capture no longer competes with presence.
- It **never forgets** and **never leaves** — institutional memory that doesn’t walk out the door.
- It **compares across every customer** to know what works.

The unlock

The data was never capturable by people — so it was never captured, so its value was never unlocked. An agent makes the impossible task routine. That's the whole opportunity: not doing what salespeople do a bit better, but doing the thing they *structurally could not do at all.*

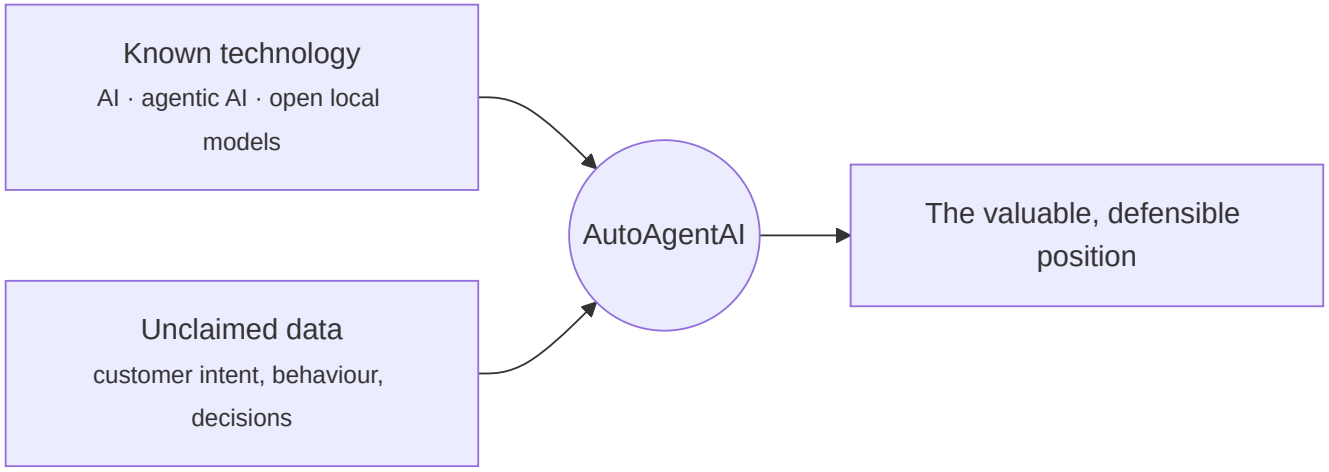
This is why the role shifts to assurance and assistance (Reinventing the Salesperson's Role) — the human does the human part, and the agent does the part no human ever could.

Capturing the Car Industry's Most Valuable Data

We are not inventing a new kind of AI. Everyone has heard of AI agents; everyone is building them; agentic AI is not a secret. What's different is **where we point it.** We are in the automotive industry, and we are using the current advances in AI, agentic AI, and open-source local models to do one specific thing nobody has done well: **capture and use the single most valuable data in the entire car industry — the customer's real intent, behaviour, and decision-making.**

> **The one idea**
>
> The technology is ordinary. The *target* is extraordinary. An AI agent — the thing everyone already knows about — aimed at the richest, least-captured data in autos, running privately on local models, closes a gap the whole industry has.

The tech is known; the application is the edge



- **AI agents** — widely understood, widely available.
- **Agentic patterns** (planning, tools, memory) — mature and documented.
- **Open-source local models** — now good enough to run the whole thing privately on-prem (Local Models Are Enough).

Individually, none of that is a moat. **Pointed at the car industry's customer behaviour data, and able to actually capture it, it is.** The moat is the data and the capture, not the model.

Why *this* data is the prize

The automotive purchase is unusually rich in signal and unusually high in stakes:

- It's a **big, considered, emotional** purchase — months of deliberation, lots of behaviour to read.
- It's **multi-channel** — web, showroom, test drive, service, finance.
- It recurs over a **long relationship** — purchase, service, trade-in, repurchase.
- The **decisions are high-value** — a single correctly-read customer is a car sold, a trade-in captured, a service relationship kept.

And almost none of it is captured today, because capturing it was impossible for the people on the floor (Why Salespeople Can't Capture This Data). The most valuable data in the industry is sitting in plain sight, uncollected.

What "capture and utilize" means here

We don't just *store* the data — the whole point is the loop from capture to use:



Everyone can run an agent. Few can *capture this data at all*, and fewer can turn it into the right action and feed it back up the chain (see Closing the Loop to the OEM). That capture → use → feedback loop is the product.

Why now, and why us

Enabler	Why it matters
Agentic AI matured	An “analyst that never sleeps” is finally buildable
Local models got good enough	The whole thing runs privately on-prem — essential for this PII-heavy data
Privacy pressure rising	Sovereign, on-prem capture is the only durable way to hold this data (Privacy)
We’re in the industry	We know where the signal is, how dealerships work, and what the OEM wants

> **The positioning**
>
> “AI agent” is table stakes. **An AI agent that captures and uses the most valuable, least-captured data in the car industry — privately, on local models, all the way up to the OEM** — is the thing. We’re not selling the hammer; we’re the ones who found the gold it’s for.

Next: just how valuable that data is — How Valuable This Data Is.

How Valuable This Data Is

Customer intent-and-behaviour data in automotive isn't "nice to have." It is, pound for pound, some of the most valuable commercial data that exists — because of what a single correct read is worth, how long the relationship lasts, and how far up the chain the signal matters. This article makes the value explicit.

> **Why it's worth so much**
>
> High transaction value × long multi-stage relationship × decisions made under uncertainty × near-zero current capture = enormous value sitting unclaimed.

Value driver 1 — the stakes per decision are huge

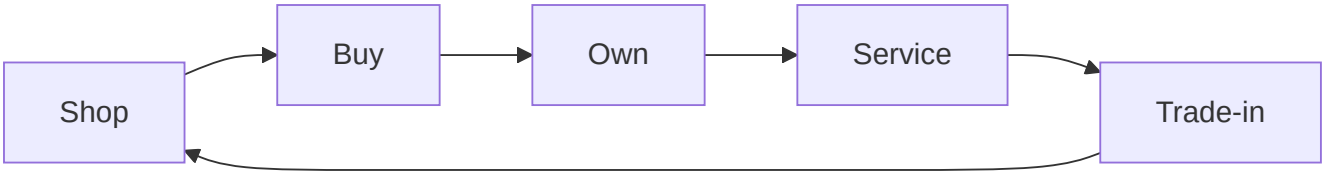
Unlike a $5 impulse buy, every read of a car customer is attached to large numbers:

Moment the data informs	Rough value at stake
A new/used vehicle sale	Tens of thousands per unit
A trade-in captured (or lost to a competitor)	A second transaction + inventory
A finance/insurance attach	High-margin add-on
A service relationship kept over years	Recurring, high-margin revenue
A repurchase at end-of-cycle	The *next* vehicle sale

Getting **one** of these right because you understood the customer pays for a lot of agent. Behaviour data here is leverage on very large transactions.

Value driver 2 — the relationship is long and multi-stage

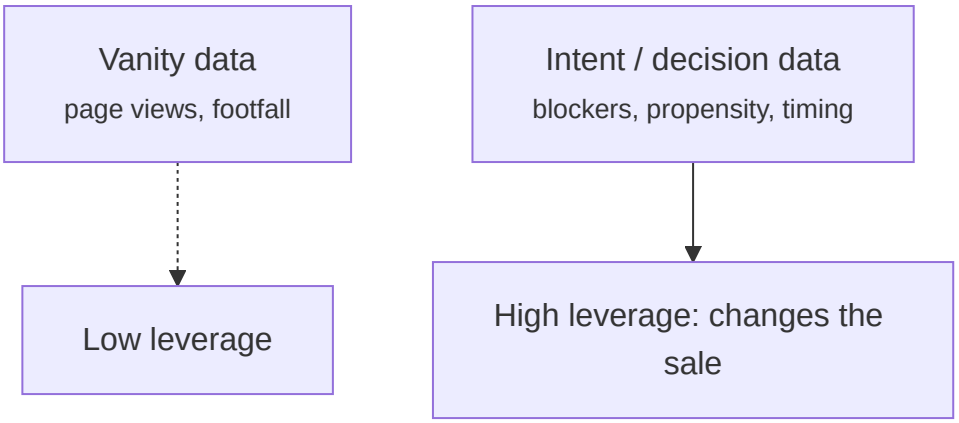
A car customer isn't a one-shot. They move through a cycle that repeats for years and decades:



Data compounds across that cycle. Knowing how someone shopped informs how you serve them; how you serve them informs whether they repurchase. Each loop makes the next read better — so the data isn't just valuable once, it's an **appreciating asset**.

Value driver 3 — it's decision data, not vanity data

Most "customer data" is traffic and vanity metrics. This is **intent and decision-making** data: *why* someone will or won't buy, what their real blocker is, what would move them. That's the data that actually changes outcomes, which is the only data worth paying for.



Value driver 4 — scarcity: almost nobody has it

Value is price × scarcity. This data is scarce precisely because it has been **impossible to capture** (Why Salespeople Can't Capture This Data). The ocean of signal exists; the captured fraction is tiny and poor. Whoever captures it well holds something competitors structurally do not.

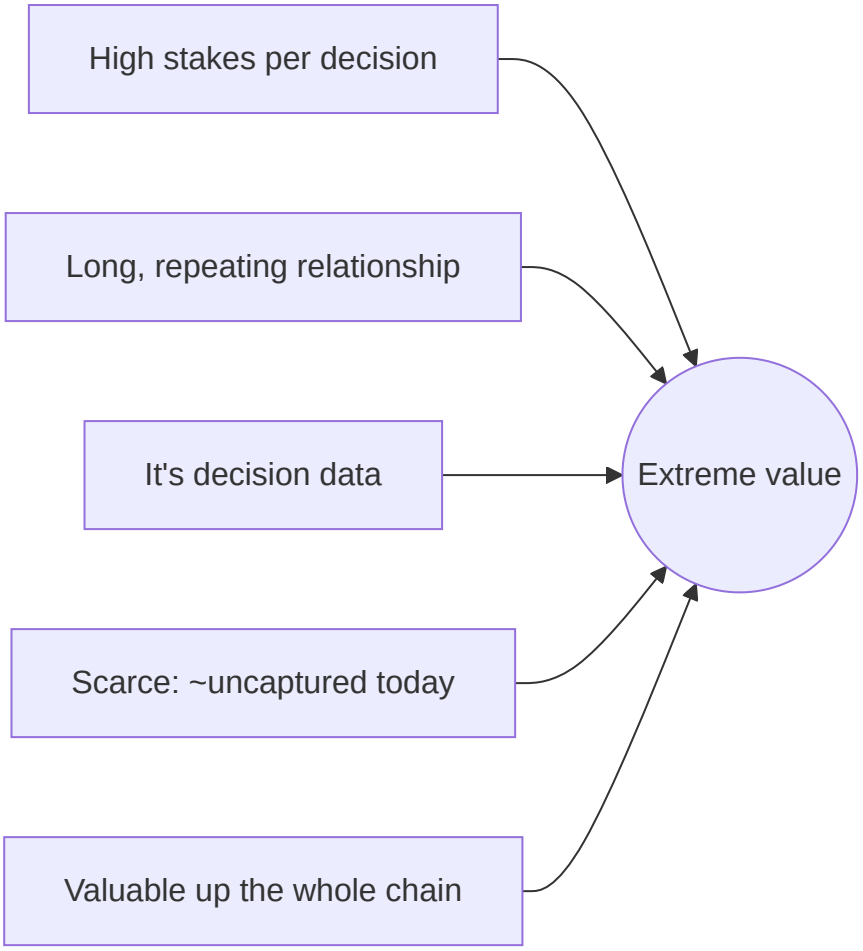
Value driver 5 — it's valuable at every level of the chain

The same data is worth money to **multiple parties**, which multiplies its value:

Level	What the data is worth to them
Salesperson	Walk up already knowing the customer's real concern
Dealership	Higher conversion, trade-in capture, retention, LTV
Dealer group	Cross-site patterns, best-practice, inventory signal
OEM / manufacturer	*What real buyers actually want and reject* — product, pricing, and marketing feedback they currently only guess at (Closing the Loop to the OEM)

Data that is valuable to the salesperson, the dealer, the group, **and** the manufacturer is far more valuable than data useful to only one of them.

Putting it together



> Big transactions, a lifelong repeating relationship, genuine decision signal, near-zero current capture, and value at every level from the salesperson to the OEM. That combination is rare. It is why this data is **up there with the most valuable commercial data anywhere** — and why capturing it is the whole game.

> **On numbers**
>
> Dollar ranges here are illustrative of *why* the value is large, not a financial model. A real value model is a Phase-1 exercise with a specific dealer's economics.

Next: the data itself, and how it closes the loop all the way to the manufacturer — Closing the Loop to the OEM.

Valuable Auto Data & Closing the Loop to the OEM

Automotive is already a data-rich industry. Manufacturers track vehicles, patterns, and behaviour at enormous scale — through connected vehicles, telematics, service records, and more. But there's a gap in that data that has stayed open: **the showroom-floor, pre-purchase, decision-making behaviour of the actual human buyer.** AutoAgentAI captures that missing piece and **closes the loop — feeding real buyer intent all the way back to the OEM, where it can change decisions.**

> **The gap in one line**
>
> OEMs know everything about the *car* and a lot about the *driver after they buy*. They know almost nothing, in real time, about *why a person did or didn't buy on the floor.* That's the loop we close.

> **What 'CV' means here**
>
> In this article **CV = connected vehicle** — the telematics, usage, and health data modern cars stream back to the manufacturer after the sale. In-dealership camera data we call **computer vision** throughout, so the two terms don't collide. The point below is that OEMs are rich in CV (post-sale) data and poor in *pre-purchase decision* data — and that's the gap.

The automotive data landscape

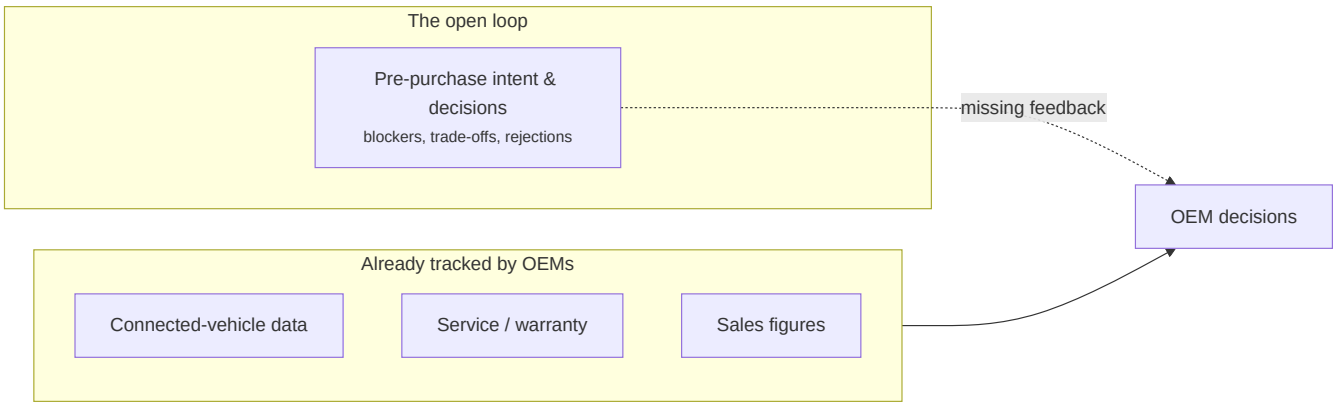
Auto data comes in several layers. Most are already well-captured; one is not.

Data layer	Example	Captured today?
Connected-vehicle (CV) / telematics	Driving patterns, usage, location, vehicle health	✅ Heavily, by OEMs
Service & warranty	Repairs, recalls, maintenance behaviour	✅ Well captured
Computer-vision / in-store sensing	Lot traffic, dwell, footfall	⚠ Partly, crudely
Transaction / finance	What sold, at what price, financed how	✅ Captured
Pre-purchase intent & decision behaviour	*Why* a buyer leaned in, hesitated, chose, or walked	❌ **Barely captured at all**

The last row is the richest in *why*, the most valuable (How Valuable This Data Is), and the least captured — because capturing it was impossible for the people on the floor (Why Salespeople Can't Capture This Data).

"They already track behaviour" — but the loop is open

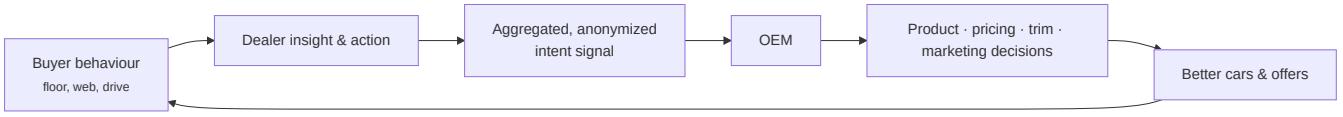
OEMs are superb at tracking the vehicle and the post-purchase driver via **connected vehicles**. What they *don't* have is a live read on the human decision that happens before the sale:



So the OEM optimizes the product and the post-sale experience with great data, and optimizes *what buyers actually want and reject* with almost none — relying on lagging sales numbers, dealer anecdote, and surveys. The loop from **buyer decision → OEM decision** is broken.

Closing the loop

AutoAgentAI captures the missing layer at the point of decision (kiosks, iPads, every channel), turns it into insight, and — with consent and in **aggregated, privacy-respecting** form — feeds it up the chain:



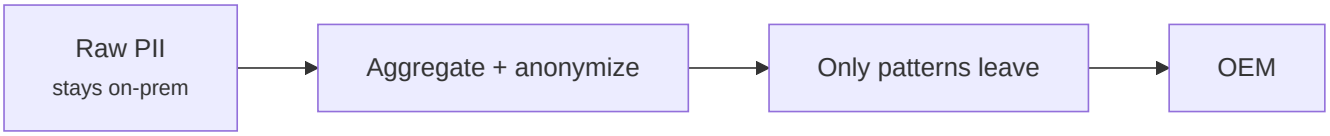
Now the OEM learns things it currently can only guess:

- **Which trims and options buyers actually weigh** — and which they reject, and why.
- **Real, live price and running-cost sensitivity** — before it shows up in sales.
- **The true blockers** that lose sales ("worried about resale," "fuel cost") at scale, in near real time.
- **What assurance closes the deal** — feeding back into messaging and product.

This complements the OEM's existing connected-vehicle and service data: they had the *car* and the *owner*; now they also have the *buyer's mind at the moment of decision.* That's the full picture.

Privacy makes the loop legitimate

Feeding data up the chain is exactly where privacy usually breaks — which is why the on-prem design matters here most. Raw PII **stays at the dealership**; only **aggregated, anonymized, consented** signal flows up. The OEM gets the *patterns*, not the people. See Privacy, PII & Future-Proofing.



Why this is a step-change, not an increment

OEMs already spend fortunes trying to infer buyer intent indirectly — clinics, focus groups, surveys, inferring it backwards from sales. AutoAgentAI gives them the real thing, measured at the moment of decision, continuously, from the actual buying population. That the loop was **impossible to close before** — and now isn't — is the subject of the next article: The Closed Loop.

> **Illustrative**
>
> Specific signals and flows are illustrative of the mechanism. What is aggregated, how, and under what consent is a privacy-and-spec design task, not a promise of a particular data-sharing arrangement.

The Closed Loop: Impossible Before, Ubiquitous Elsewhere

Closing the loop — from a buyer's real decision behaviour back to the people who make the cars — was not just *hard* before. It was **impossible**. The pieces didn't exist. Now they do, all at once, and that's the opening. This article is about why it couldn't be done, why it can be now, how valuable closed loops are in general, and how other industries already prove it (including the clumsy attempts automotive has made so far).

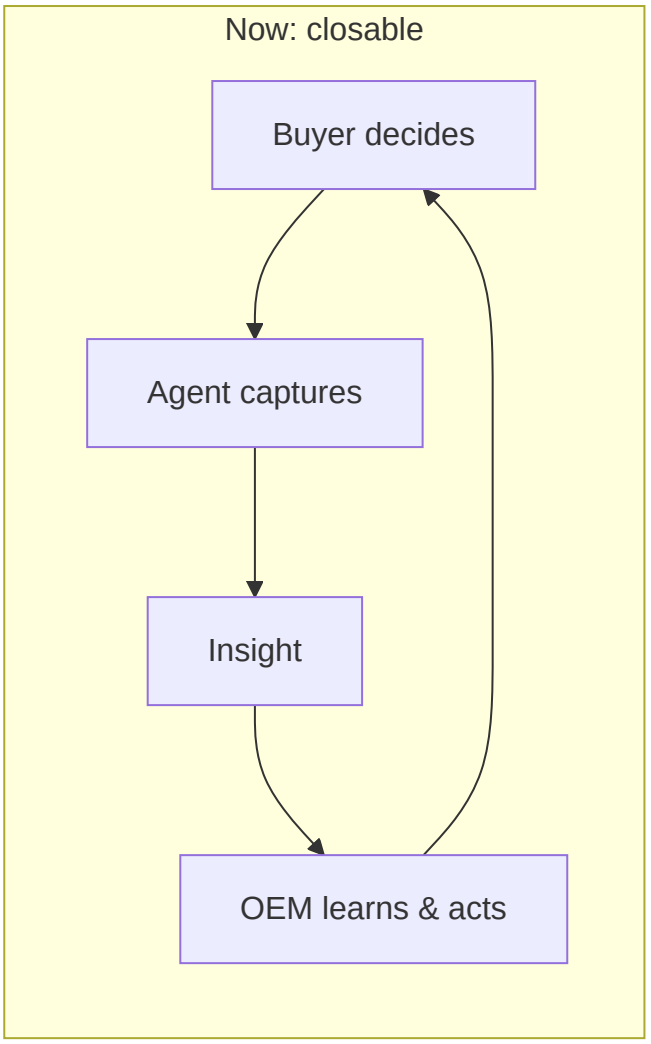
> **The arc**
>
> Every great data business is a closed loop: observe → learn → act → observe better. Automotive's buyer-decision loop stayed open because no tool could close it. The tools finally exist — so the last great open loop is closable.

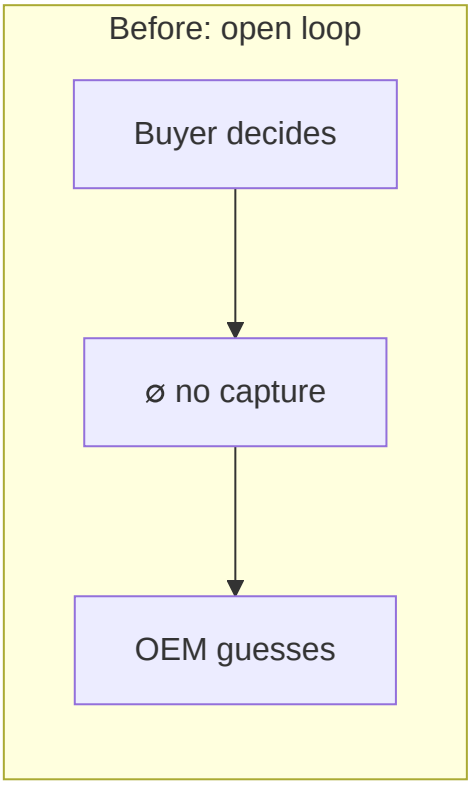
Why it was impossible before

Four things all had to be true at once, and until recently none were:

Requirement	Why it was missing
Capture at the point of decision	No human could capture it (here's why); no system was in all channels
Make sense of messy, human signal	Reading intent from notes, hesitation, behaviour needed AI that didn't exist affordably
Do it privately	This is PII-heavy; moving it to the cloud was a non-starter — and local models weren't good enough (now they are)
Act and feed back autonomously	Required agentic orchestration that's only just matured

Miss any one and the loop stays open. For decades, all four were missing. The loop wasn't closed because it *couldn't* be.


Now: closable
Buyer decides
Agent captures
Insight
OEM learns & acts


Before: open loop
Buyer decides
⌀ no capture
OEM guesses

Why it's possible now

The same four requirements, flipped:

1. **Kiosks + iPads in every channel** capture decision behaviour as a by-product of being useful (On-Prem AI & Hardware).
2. **Agentic AI** reads the messy human signal and reasons over it.
3. **Open local models** make it private and affordable — the data never leaves.
4. **The agent acts and feeds back** continuously, closing the loop end to end.

For the first time, all four are true together. That simultaneity *is* the opportunity.

Closed loops are how every data giant wins

This pattern isn't exotic — it's the engine behind the most valuable companies in the world. They all close a behaviour → action → better-behaviour loop:

Industry / player	The closed loop they run
Streaming	What you watch → recommendations → you watch more → better recommendations
E-commerce	What you browse/buy → personalization → more conversion → sharper model
Social / ads	Engagement → feed & ad targeting → more engagement → more data
Search	Clicks → ranking → better results → more clicks
Rideshare / delivery	Demand signal → pricing & dispatch → smoother service → more demand

In each, the loop compounds: more data → better action → more data. The winners aren't the ones with the fanciest model; they're the ones who **closed the loop** on valuable behaviour first. Automotive's buyer-decision loop is one of the last big ones still open.

Automotive's clumsy attempts so far

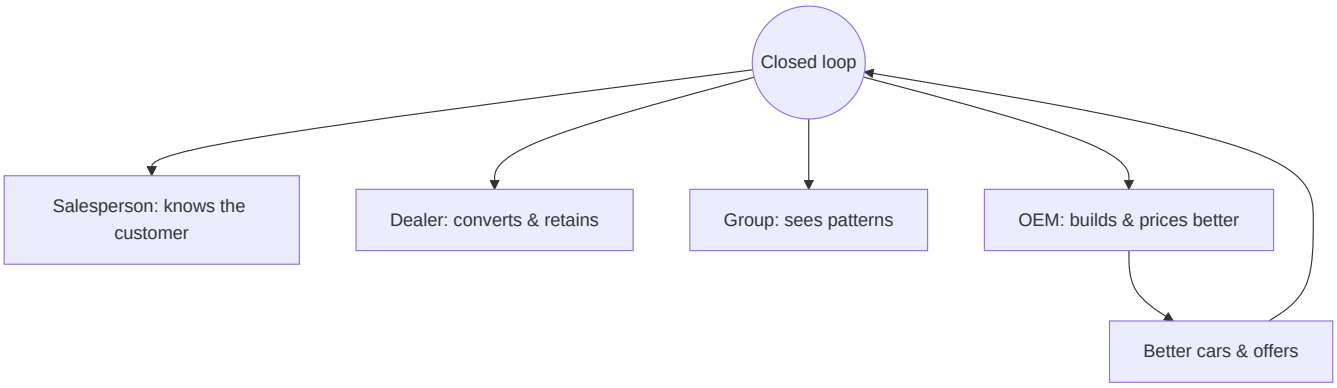
The industry *knows* it wants this data — its attempts to get it show the hunger and the failure:

Attempt	Why it falls short
Feedback surveys (often with prizes)	Low response, self-selected, after the fact, memory-distorted; a prize bribes a form, it doesn't capture the decision
"How was your visit?" emails / CSI scores	Measure satisfaction, not intent; gameable; miss the people who walked
Mystery shoppers	A tiny, artificial sample
Focus groups & clinics	Expensive, infrequent, not the real buying population
Inferring from sales figures	Lagging, and silent on *why* — especially on the sales you *lost*

Every one of these is a workaround for not being able to capture the real thing at the moment it happens. They're proof the demand is there — and proof the problem was unsolved. A survey with a prize draw is what you do when you *can't* close the loop. AutoAgentAI is what you do when you can.

The ubiquity value

Because the same captured data is valuable at **every** level — salesperson, dealer, group, and OEM (How Valuable This Data Is) — closing this loop isn't one feature for one user. It's an upgrade to the whole chain at once:

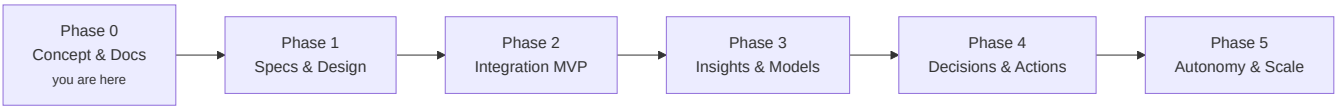


The possibility

Every industry that closed its behaviour loop was transformed by it. Automotive couldn't — until now. The possibilities aren't incremental: a continuously learning chain from the showroom floor to the factory, where what buyers actually want reaches the people who decide what gets built. That loop was impossible yesterday and is buildable today. That's the opportunity in one sentence.

Roadmap

A phased path from **concept and documentation** (where the project is now) to a validated product. Phases are about *proving the next risky thing*, not calendar dates.



Phase 0 — Concept & documentation ✅ *(current)*

Goal: get the thinking organized, shared, and hosted.

- [x] Repository created, documentation-first.
- [x] Concept, vision, and the five-pillar architecture documented.
- [x] Shared vocabulary (glossary).
- [x] Diagrams (Mermaid) + figure specs for richer visuals.
- [x] Hosted docs site (GitHub Pages via MkDocs Material).

Phase 1 — Specifications & design

Goal: turn concept into buildable specs. De-risk the *what* and *how*.

- [ ] Choose the beachhead use case and target customer segment.
- [ ] Physical data model + schema for the unified profile.
- [ ] Technical architecture decision records (stack, model providers, datastores).
- [ ] Define v1 action catalog and default guardrails.
- [ ] Spec the first behaviour models (likely RFM, churn/purchase propensity, behavioural clustering).
- [ ] API and integration contracts for the first 1–2 data sources.

→ Specs live under Specifications.

Phase 2 — Customer Integration MVP

Goal: prove we can integrate real customer data and resolve identities well.

- [ ] Ingest from the first sources (e.g. commerce + one behavioural source).
- [ ] Identity resolution with confidence scoring.
- [ ] Unified profiles materialized and queryable.
- [ ] Consent/privacy enforcement baseline.

Phase 3 — Insights & behaviour models

Goal: prove we can produce genuinely useful, grounded insights.

- [ ] First behaviour models trained and serving.
- [ ] Insights Engine producing validated, ranked insights.
- [ ] Plain-language “ask the agent” (pull mode).
- [ ] Human evaluation: are the insights trusted and non-obvious?

Phase 4 — Decisions & actions

Goal: close the last mile — insight → action → measured outcome.

- [ ] Next-best-action and action catalog.
- [ ] First outbound action integration (e.g. trigger a campaign / create a task).
- [ ] Experimentation + attribution framework (holdouts).
- [ ] Human-in-the-loop approval console.

Phase 5 — Autonomy & scale

Goal: earn and expand autonomy; make it robust and multi-tenant.

- ☐ Graduated autonomy levels with per-action guardrails.
- ☐ Continuous agent loop with memory and learning from outcomes.
- ☐ Observability, audit, and safety hardening.
- ☐ Scale to more sources, more customers, more businesses.

Risks to retire (in rough priority)

Risk	Phase it's tested
Can we integrate & resolve messy real data?	2
Are the insights actually useful and trusted?	3
Does acting on them move the metric (causally)?	4
Can the agent act autonomously *and* safely?	5
Does it generalize across businesses?	5

> **This roadmap is a draft**
>
> Sequencing and scope will change as specs are written and the beachhead use case is chosen. Phase 0's job is to make that conversation possible.

Specifications

> **Placeholder — specs begin in Phase 1**
>
> This section will hold the **formal, buildable specifications** for AutoAgentAI. It is intentionally a stub today: the concept docs (Phases 0) come first so the specs have a foundation to build on. See the roadmap.

How specs will be organized

Each spec is a focused document with a consistent shape so they're easy to read, review, and implement against:

Spec area	Scope
Data model spec	Physical schema for unified profiles, events, features.
Integration spec	Per-source ingestion, identity-resolution policy, consent enforcement.
Behaviour-model specs	Each model: inputs, method, outputs, training/serving, metrics.
Insights spec	Insight object schema, generation methods, validation bar, ranking.
Decision spec	Action catalog, next-best-action logic, guardrails, attribution.
Agent spec	Loop, planning, tools, memory, autonomy levels, safety.
API specs	Contracts between pillars and with external systems.

Spec document template (to be used)

Every spec should contain:

1. **Purpose & scope** — what this spec covers and explicitly does not.
2. **Context** — where it sits in the architecture.
3. **Requirements** — functional and non-functional, numbered.
4. **Design** — the chosen approach, with diagrams.
5. **Interfaces / schema** — concrete contracts.
6. **Decisions & trade-offs** — key choices and why (ADR-style).
7. **Open questions** — what's still undecided.
8. **Acceptance criteria** — how we know it's correctly built.

Architecture Decision Records (ADRs)

Load-bearing technical choices (stack, model providers, datastore, agent framework) will be captured as short ADRs so the *why* behind each decision is preserved. An ADR records: context, the decision, the alternatives considered, and the consequences.

When Phase 1 begins, replace this stub with the real specs and expand the navigation in `mkdocs.yml` to list them.

Figure Specs

Two kinds of visuals are used in these docs:

1. **Mermaid diagrams** — written as code throughout the docs. They render automatically on GitHub *and* on the hosted site, and are editable as text. No image generation needed. The rest of this page lists where they live.
2. **Rich conceptual visuals** — polished illustrations Mermaid can't produce (hero images, layered concept art, marketing-grade figures). These are **specified below as generator-ready prompts** so you can produce them later with an image tool and drop the files into `docs/assets/diagrams/`.

Part A — Mermaid diagrams already in the docs

Diagram	Location	Shows
Five pillars	`README.md`, `index.md`	The pillar pipeline + agent
Product-in-one-picture	`index.md`	Sources → integration → agent → actions
Fragmentation problem	`problem-motivation.md`	Why data is siloed
Pillar pipeline (detailed)	`architecture.md`	All five pillars internally
Logical layers	`architecture.md`	Source → data → intelligence → agent → experience
End-to-end sequence	`architecture.md`	The full request/response + feedback
Integration flow + ERD	`customer-integration.md`	Ingest→resolve→enrich; conceptual entities
Insight generation	`insights-engine.md`	Methods → candidates → rank
Behaviour segmentation quadrant	`behaviour-models.md`	Value vs. engagement
Lifecycle state machine	`behaviour-models.md`	Prospect → … → churned/reactivated
Insight→decision→outcome	`decision-making.md`	The last-mile loop
Agent loop + autonomy ladder	`agent-loop.md`	Perceive→reason→act→learn; L0–L4
Use-case flows	`use-cases.md`	Per-scenario arcs
Roadmap phases	`roadmap.md`	Phase 0→5

To edit any of these, find the ```` ```mermaid ```` block in the named file. Preview with `mkdocs serve` or directly on GitHub.

Part B — Rich visuals to generate (prompts)

Each entry gives a **suggested filename**, **where it would go**, and a **generator prompt**. Style guidance is shared below so the set looks consistent.

Shared style guide

> Modern, clean, professional SaaS aesthetic. Indigo-to-deep-purple accent palette on a light background (and a dark-mode variant where noted). Flat vector / subtle depth, generous whitespace, rounded shapes, thin consistent linework. No stock-photo people, no clutter, no fake chart numbers. Legible labels. 16:9 unless stated.

Fig 1 — Hero / concept image

- **File:** `hero.png` · **Use:** top of `README.md` / site home.

- **Prompt:** > A clean, modern hero illustration for an AI product called “AutoAgentAI.” Center: > a friendly abstract AI agent represented as a glowing node/orb made of connected > lines. Flowing into it from the left, several labeled data streams (transactions, > web events, CRM, support, reviews) merging into one. Flowing out to the right, > three labeled outputs: “Insights,” “Decisions,” “Actions.” Indigo-to-purple > gradient accents on a light background, flat vector style, lots of whitespace, > subtle glow. 16:9.

Fig 2 — The five pillars (polished)

- **File:** `pillars.png` · **Use:** `architecture.md` header (alongside the Mermaid version).
- **Prompt:** > A polished horizontal diagram of five connected pillars for a customer-intelligence > platform. Four pillars in a left-to-right pipeline labeled “Customer Integration,” > “Behaviour Models,” “Insights Engine,” “Decision Intelligence,” each as a rounded > card with a simple icon. A fifth element, “The AutoAgent,” sits above as an orb > connected by dotted lines to all four, labeled “orchestrates.” Indigo/purple on > light, flat vector, clean icons, thin connectors with arrowheads. 16:9.

Fig 3 — The autonomous agent loop (polished)

- **File:** `agent-loop.png` · **Use:** `agent-loop.md` header.
- **Prompt:** > A circular loop diagram with four labeled stages arranged clockwise: “Perceive,” > “Reason & Plan,” “Act,” “Learn,” connected by curved arrows forming a continuous > cycle. In the center, a small AI agent orb. Each stage has a minimal icon (eye, > brain/gears, lightning/hand, upward arrow). Indigo/purple accents on light > background, flat vector, clean. Square (1:1).

Fig 4 — Customer 360 / unified profile

- **File:** `customer-360.png` · **Use:** `customer-integration.md`.
- **Prompt:** > An illustration of identity resolution: on the left, five scattered, mismatched > record cards (POS receipt, CRM contact, anonymous web-device icon, support ticket, > email contact), each with a small question mark. Arrows converge to the right into > a single clean “unified customer profile” card showing combined attributes. > Convey “many fragments → one resolved view.” Indigo/purple on light, flat vector, > no real personal data. 16:9.

Fig 5 — Insight anatomy card

- **File:** `insight-anatomy.png` · **Use:** `insights-engine.md`.
- **Prompt:** > A single clean “insight card” UI mockup for an AI analytics product. It shows, > top to bottom: a plain-language Statement, an Evidence row, a Confidence meter, a > Scope (“applies to N customers”), an Expected impact figure, and a “Suggested > action” button. Minimal, modern dashboard style, indigo/purple accents, light and > dark variants. 4:3.

Fig 6 — Autonomy ladder

- **File:** `autonomy-ladder.png` · **Use:** `agent-loop.md` (guardrails section).
- **Prompt:** > A five-step ascending staircase, each step a labeled level: “L0 Observe,” > “L1 Recommend,” “L2 Act with approval,” “L3 Act within limits,” “L4 Autonomous.” > A small agent orb climbs the steps; a dotted “trust / track record” arrow runs > upward alongside. Indigo/purple gradient darkening as it ascends, light > background, flat vector, clean labels. 16:9.

Fig 7 — Roadmap timeline

- **File:** `roadmap.png` · **Use:** `roadmap.md` header.
- **Prompt:** > A clean horizontal roadmap with six milestones labeled “Phase 0 Concept & Docs,” > “Phase 1 Specs,” “Phase 2 Integration,” “Phase 3 Insights,” “Phase 4 Decisions,” > “Phase 5 Autonomy.” Phase 0 marked as current/done with a checkmark. Each milestone > a node on a path with a small icon. Indigo/purple on light, flat vector. 16:9.

How to add a generated image to the docs

1. Generate the image from the prompt and save it into `docs/assets/diagrams/` (e.g. `docs/assets/diagrams/hero.png`).
2. Reference it in the relevant markdown page, for example in `index.md`:

   ```
   ![AutoAgentAI concept](assets/diagrams/hero.png)
   ```

   (Paths are relative to the markdown file. From `docs/index.md`, use `assets/diagrams/hero.png`; from a page in `docs/specs/`, use `../assets/diagrams/hero.png`.)
3. Commit and push — the site rebuilds automatically.

> **You don't have to generate anything to ship**
>
> The Mermaid diagrams already carry every page. The Part B visuals are polish for when you want a more designed look — generate them on your own schedule.

Part C — Deployment & perspective visuals (prompts)

Same shared style guide as Part B. These back the automotive, on-prem, and audience pages.

Fig 8 — The four deployment tiers

- **File:** `tiers.png` · **Use:** `deployment/on-prem-ai-hardware.md`, `audiences/investor-pitch.md`.
- **Prompt:** > A clean horizontal "ladder" of four ascending tiers labeled "Tier 0 Cloud," > "Tier 1 Local Essentials," "Tier 2 Local Pro," "Tier 3 Local Enterprise." Tier 0 > sits lower with a small cloud icon and a subtle warning tint ("data leaves > premises"); Tiers 1–3 ascend with a padlock and on-prem server icons growing from > a single box to a rack, each tagged "data stays on-site." Indigo→purple gradient > darkening as it ascends, light background, flat vector. 16:9.

Fig 9 — On-prem topology: kiosks & iPad nodes

- **File:** `topology.png` · **Use:** `deployment/on-prem-ai-hardware.md`, `audiences/technical-overview.md`.
- **Prompt:** > A hub-and-spoke diagram. Center: a single on-prem server box labeled "On-prem > node — local AI models + customer data + agent," drawn with a padlock. Spokes out > to several thin devices: two large self-serve kiosks and three iPads, each labeled > "node (thin client)." One dotted arrow leaves the hub labeled "only aggregated, > consented signal → OEM." Emphasize that intelligence/data stay central, devices > are thin. Indigo/purple on light, flat vector. 16:9.

Fig 10 — Closing the loop to the OEM

- **File:** `closed-loop.png` · **Use:** `perspectives/automotive-data-closing-the-loop.md`, `perspectives/the-closed-loop.md`.
- **Prompt:** > A large circular loop connecting four labeled nodes clockwise: "Buyer decides (on > the floor)," "Agent captures & acts," "Aggregated intent signal," "OEM builds & > prices better," returning to the buyer via "Better cars & offers." Raw-data icon > near the buyer marked "stays on-prem," and the signal going up marked "anonymized." > Convey a previously-broken loop now closed. Indigo/purple on light, flat vector. > Square or 16:9.

Fig 11 — Raw data → action funnel

- **File:** `data-to-action.png` · **Use:** `perspectives/raw-data-to-action.md`, home.
- **Prompt:** > A clean left-to-right funnel/pipeline with five stages: "Raw data (scattered)," → > "Useful data (one profile)," → "Insight," → "Action," → "Outcome," with a thin > dotted "feedback" arrow curving from Outcome back to Raw data. Scattered dots on > the left gradually resolve into a single clean figure on the right. Indigo/purple > on light, flat vector. 16:9.

Fig 12 — "Why now": three converging curves

- **File:** `why-now.png` · **Use:** `audiences/investor-pitch.md`.
- **Prompt:** > Three labeled trend lines rising and converging at a single highlighted point > marked "The opening / now": "Agentic AI capability," "Open local-model quality," > and "Privacy-law pressure." Minimal chart aesthetic, no fake numbers, clean axes, > the convergence point glowing. Indigo/purple accents on light. 16:9.